Money Laundering and the Privacy Design of Central Bank Digital Currency*

Zijian Wang†

Wilfrid Laurier University

March 2023

Abstract

This paper studies the implications of money laundering for the optimal design of central bank digital currency (CBDC). I build a general equilibrium framework to explicitly allow money laundering by agents and income audits by a government. I find that as long as CBDC offers less anonymity than cash, introducing CBDC will decrease money laundering. However, if CBDC is "*cash-like*" in the sense that it still offers a relatively high level of anonymity but a low interest rate, then introducing CBDC will decrease the output from not only agents who launder money but also agents who do not. If CBDC is instead "*deposit-like*" in the sense that it offers low anonymity but a high interest rate, then introducing CBDC can increase aggregate welfare without lowering output. Furthermore, introducing deposit-like CBDC needs not increase the funding costs of banks or decrease bank lending.

JEL Codes: K14, E51, E52, E58

Keywords: Money laundering, Central bank digital currency, Monetary policy, Banking

*This paper subsumes an earlier version titled "Tax Compliance, Payment Choice, and Central Bank Digital Currency". I am indebted to Stephen Williamson and Lucas Herrenbrueck for their guidance and encouragement. I also thank editor Marco Bassetto and the anonymous referees whose suggestions greatly improved this paper. I benefited from discussions with Lukas Altermatt, Jeremie Banet, Aleksander Berentsen, Zach Bethune, Daniel Brou, Elizabeth Caucutt, Jonathan Chiu, Michael Choi, Edda Claus, Simona Cociuba, Huberto Ennis, Athanasios Geromichalos, Pedro Gomis-Porqueras, Francisco Gonzalez, Juan Carlos Hatchondo, Thorsten Koeppl, Ioannis Kospentaris, Sergio Ocampo, Luba Petersen, Roberto Pinheiro, Stanislav Rabinovich, Baxter Robinson, Guillaume Rocheteau, Brenda Samaniego, Christos Shiamptanis, Mario Silva, Amy Hongfei Sun, and Shengxing Zhang as well as seminar participants at various conferences and seminars. All errors are mine.

†Email: zijianwang@wlu.ca

Electronic copy available at: https://ssrn.com/abstract=4394190

1 Introduction

Central bank digital currency (CBDC), which is a digital form of central bank money, has attracted worldwide attention in recent years. A survey by the Bank for International Settlements finds that currently, more than 80% of central banks are actively researching CBDC (Boar et al., 2020). Although there is not yet a consensus on how CBDC should be designed, many central bankers argue that the potential use of CBDC in illicit activities, such as money laundering, should be taken into account.[1] The goal of this paper is to study how CBDC should be designed when money laundering is a concern in the economy, and CBDC may compete with other anonymous payment instruments such as cash. I ask how the introduction of CBDC affects the choice of payment instruments, money laundering, and aggregate welfare.

To answer these questions, I develop a general equilibrium framework based on the model of Lagos and Wright (2005). A key feature of the framework is that the problems of money laundering and government income audits are studied jointly with the problem of payment choice. Specifically, in the economy, there is a (legal) consumption good sold by sellers, who can choose the payment instrument(s) they accept. Sellers may also launder money for criminals by disguising illegal income as part of sellers' legal revenue so long as both incomes are received in the same payment instrument(s). At the end of each period, the government may audit sellers. After an audit, the probability of a seller's money laundering activities being discovered depends on the payment instrument(s) in which legal and illegal incomes are received. If a seller is found laundering money, the government can confiscate the income discovered.

As a benchmark, I assume cash and bank deposits are the only payment instruments. I find that in equilibrium, some sellers choose to launder money. As long as it is sufficiently easier to hide cash compared to bank deposits, sellers who engage in money laundering accept cash despite its lower rate of return. The main results of this paper concern the optimal design of CBDC when cash facilitates money laundering. I define CBDC to be another payment instrument (in addition to cash) that is issued by the central bank. CBDC can be different from cash and bank deposits depending on two design choices of the central bank: the interest rate on CBDC and how much anonymity CBDC offers.[2] In the context of money laundering, the degree of anonymity of CBDC determines the probability of money laundering being discovered after a government audit if a seller's incomes are received in CBDC.

I find that as long as CBDC offers less anonymity than cash, then introducing CBDC will decrease money laundering. However, if CBDC is "*cash-like*" in the sense that it still offers a

[1] See for example Lagarde (2018), Powell (2020), Bank of England (2020), and Bank of Canada (2020). See also Bank of Canada et al. (2020) for an overview of central banks' motivations for introducing CBDC.

[2] The anonymity of CBDC can indeed be a choice of the central bank. For example, Darbha and Arora (2020) show the types of cryptographic techniques and operational arrangements that can be used to achieve fine-grained privacy designs. ESCB (2019) demonstrates how CBDC can offer anonymity through "anonymity vouchers".

Electronic copy available at: https://ssrn.com/abstract=4394190

relatively high level of anonymity but low interest rate, then introducing CBDC will decrease the output from sellers who launder money. Specifically, in equilibrium, sellers who launder money substitute cash with CBDC because the latter pays a positive interest rate. However, because CBDC offers less anonymity than cash, the visibility of sellers' assets increases. This gives the government more incentive to conduct audits. The higher audit probability increases the risk of money laundering being discovered. Consequently, fewer sellers choose to launder money, and sellers who launder money produce less and receive smaller payments.

Perhaps more interestingly, the output from sellers who do not engage in money laundering also decreases following the introduction of a cash-like CBDC. In equilibrium, because the interest rate on CBDC is low, sellers who do not launder money prefer to accept bank deposits. The decrease in money laundering prompts more sellers to switch to accepting bank deposits, which increases the demand for deposits and drives down the deposit rate. A lower deposit rate decreases the income of sellers who do not launder money and reduces their output. I also find that increasing the interest rate on cash-like CBDC will further reduce money laundering and worsen the shortage of bank deposits. As a result, the output from sellers who do not launder money is decreasing in the interest rate on cash-like CBDC.

It is, however, possible to design CBDC in a way that reduces money laundering while also avoiding a shortage of bank deposits. Specifically, the interest rate on CBDC should be set sufficiently high so that sellers who do not launder money are willing to accept CBDC. Meanwhile, CBDC should offer the least amount of anonymity such that sellers who launder money are also willing to accept CBDC. I find that in such an equilibrium, sellers who do not launder money are indifferent between accepting deposits and CBDC, while sellers who launder money prefer CBDC to cash. I refer to this type of CBDC as "*deposit-like*". Similar to cash-like CBDC, introducing deposit-like CBDC decreases money laundering and increases the demand for bank deposits. However, because CBDC offers a high return, sellers who do not launder money are willing to substitute bank deposits with CBDC. This avoids a shortage of deposits and a decrease in the deposit rate. Hence, introducing deposit-like CBDC does not decrease sellers' output. Furthermore, aggregate welfare is improved due to the reductions in money laundering and the real resources devoted to government audits.

It has been argued that CBDC may compete with bank deposits and increase the funding costs of private banks (Garcia et al., 2020; Bank of England, 2020; Bank of Canada, 2020). Indeed, Keister and Sanches (2023) show that while CBDC can promote efficiency in exchange, it also crowds out bank deposits and decreases investment. The same trade-off also exists in this paper when the interest rate on CBDC is high. However, if the central bank sets the interest rate on CBDC to be the same as the deposit rate, then introducing such a CBDC will have no impact on the loan rate and the deposit rate. As a result, bank loans and the funding cost of banks will be unaffected. Nevertheless, CBDC can still have real effects on the economy: when

Electronic copy available at: https://ssrn.com/abstract=4394190

CBDC's anonymity is chosen optimally, aggregate welfare is increased due to the decreases in money laundering and government audit costs.

This paper is closely related to the recent literature that studies payment-related privacy issues. Kahn et al. (2005) study cash's advantage in preserving privacy when compared with credit. Garratt and Van Oordt (2021) show that individuals tend not to protect their payment privacy to the socially optimal level. Garratt and Lee (2021) and Ahnert et al. (2022) assume payment data contains information that can be used for price discrimination and show how privacy-preserving CBDC may improve welfare. Kang (2021) and Chiu and Koeppl (2022) focus on the incentive of platforms to issue digital currencies when payment data can be used to estimate consumers' preferences. I complement the existing literature by focusing on nefarious uses of the privacy feature of payment instruments and deriving the optimal amount of anonymity CBDC should offer. I show that a welfare-maximizing CBDC must offer less anonymity than cash but also enough anonymity to attract sellers who launder money.

This paper is also related to the literature that studies the informal economy in the Lagos and Wright (2005) framework. Sanches and Williamson (2010) assume that cash can be stolen. They find that some inflation is optimal, since it lowers the value of stolen cash and deters theft. Gomis-Porqueras et al. (2014) and Aït Lahcen (2020) assume that cash transactions are not observable to the government and therefore can be used for tax evasion. They find a negative relationship between tax evasion and inflation. Similar to these papers, I find that payment instruments' rates of return (e.g., inflation; deposit rate; interest rate on CBDC) affect the degree of money laundering and the government's incentive to conduct audits. Other papers that also study the relationship between payment and tax evasion include Aruoba (2021), Bajaj and Damodaran (2022), and Kwon et al. (2022). Compared to the literature, my main contribution is that I endogenize the government's audit effort, which proves to be a crucial channel through which money laundering affects the optimal design of CBDC.

Finally, this paper is related to the emerging literature on CBDC. Chiu et al. (2022) and Andolfatto (2021) assume that banks have market power and find that CBDC improves efficiency in the banking sector. Williamson (2022) considers an environment where banks are subject to limited commitment and finds that introducing CBDC improves welfare. Regarding financial stability, Fernández-Villaverde et al. (2021) find that while CBDC can deter bank runs, it will also attract deposits from commercial banks. Schilling et al. (2020) show that it is impossible for the central bank to achieve efficiency, financial stability, and price stability simultaneously with CBDC. Keister and Monnet (2020) show that CBDC can offer real-time information to the central bank and regulators, while Williamson (2021) finds that CBDC can reduce the damages resulted from a banking panic as it is less disruptive of retail payments.[3]

[3]For more discussions on CBDC, see Barrdear and Kumhof (2016), Bordo and Levin (2017), Berentsen and Schar (2018), Ricks et al. (2018), Brunnermeier and Niepelt (2019), Kahn et al. (2020), Davoodalhosseini (2021), Dong and Xiao (2021), and Agur et al. (2022).

Electronic copy available at: https://ssrn.com/abstract=4394190

2 Model Environment

The model builds on the Lagos and Wright (2005) framework. Time is discrete, and each period is divided into two subperiods: the decentralized market (DM) and the centralized market (CM). There is measure one of infinitely-lived buyers and measure $\alpha > 1$ of infinitely-lived sellers. In the DM, buyers consume a DM good that can only be produced by sellers. In the CM, sellers consume a CM good that can be only produced by buyers. The CM good also serves as the numéraire. A buyer's instantaneous utility is given by

$$u(x_t) - l_t, \tag{2.1}$$

where x_t is the consumption of DM good, and l_t is the labor supplied in the CM. One unit of labor can be turned into one unit of CM good. I assume $u'(x) > 0$, $u''(x) < 0$, $u(0) = 0$, $u'(\infty) = 0$, $u'(0) = \infty$, and $-xu''(x)/u'(x) < 1$. A seller's instantaneous utility is given by

$$-h_t + X_t, \tag{2.2}$$

where h_t is the labor supplied in the DM, and X_t is the consumption of CM good. I assume one unit of labor in the DM can be turned into one unit of DM good. All agents discount future utility using $\beta \in (0, 1)$. Neither good can be carried across periods.

In the DM, buyers and sellers trade the DM good, and the terms of trade are determined via price posting. Specifically, the market in the DM consists of different submarkets. Each submarket is identified by its terms of trade, (x, p), where x represents the quantity of DM good a seller in this submarket offers to produce, and $p \equiv (m, c, d)$ represents the payment that the seller expects to receive, with m, c, and d denoting payments in cash, central bank digital currency (CBDC), and bank deposits, respectively. The terms of trade for the DM of period t are posted in the CM of period $t - 1$, and sellers can commit to the terms of trade they post. Buyers observe all terms of trade before they decide which market they will visit and how much cash, CBDC, and bank deposits they will carry. Let $n(x, p)$ denote the buyer-to-seller ratio in each submarket. I assume that a buyer meets a seller with probability $\min\{1, 1/n(x, p)\}$, and a seller meets a buyer with probability $\min\{1, n(x, p)\}$.

In addition to selling the DM good (which is a legal good), sellers may use their businesses to engage in money laundering. Specifically, sellers can accept illegal income from criminals in the CM, disguise it as part of the legal income in the DM, and then return the income to the criminals in the next CM. I assume that the amount of illegal income a seller can launder in a particular payment instrument (e.g., cash, bank deposits, and CBDC) is proportional to the seller's legal income in the same payment instrument.[4] To highlight the mechanisms behind the optimal privacy design of CBDC, in the main text of this paper, I abstract from the market

[4] According to Wikipedia, money laundering typically involves legal enterprises mixing illegal income with legal income. In such cases, the higher a seller's legal revenue is, the more illegal income can be laundered.

Electronic copy available at: https://ssrn.com/abstract=4394190

for money laundering services and activities that generate illegal income. I assume instead that the supply of illegal income for laundering is perfectly elastic, and that sellers receive a payoff $\mathcal{P}$ per unit of money laundered and incur a unit cost $\mathcal{C}$. In Appendix D.1, I model explicitly the markets for illegal goods and money laundering services to endogenize the source of illegal income as well as the payoff $\mathcal{P}$. I show that the main results remain unchanged.

I assume that the laundered assets are used to fund more criminal activities in the CM, which generate a welfare cost proportional to the laundered assets. The welfare cost can be avoided if the assets of sellers who launder money are discovered and confiscated through government audits. Specifically, the government may conduct audits at the end of the DM, and each audit costs $\tilde{C}$ units of numéraire.[5] After an audit, the government discovers a seller's assets with probability one if they are in the form of bank deposits. However, cash assets are only observed with probability $\rho^m \in (0, 1)$. As for CBDC, I consider various regimes where the probability of detection is higher or lower compared to other payment instruments.

In the CM, a perfectly competitive market opens for agents to trade the CM good. There is also measure one of bankers and entrepreneurs who are active. Bankers derive linear utility from the CM good, and they can produce the CM good with the same linear technology that buyers use. I assume bankers have limited commitment and can choose to default on their deposit liabilities. Therefore, bank deposits must be backed by other assets such as cash, CBDC, government bonds, and loans to entrepreneurs. I assume entrepreneurs are born in each CM with a one-period project that takes the CM good as input (denoted by k) and yields the CM good in the CM of the next period as output (denoted by $f(k)$, where $f'(k) > 0$, $f''(k) < 0$, $f'(0) = \infty$, and $f'(\infty) = 0$). Entrepreneurs derive linear utility from consuming the CM good in the second CM of their lives before they die and are replaced with a new set of entrepreneurs. Entrepreneurs are born without any funds, so they must borrow from bankers. Finally, the loan and deposit markets are assumed to be perfectly competitive.[6]

The government in the model also includes a fiscal authority that issues one-period nominal government bonds, which are traded in a competitive bond market in the CM. Let $\mathbb{B}_t^c$ be the government bonds held by the central bank. Denote $\mathbb{M}_t$ and $\mathbb{C}_t$ as the total supply of cash and CBDC in period t. Let the nominal bond rate and the nominal interest rate on CBDC be R^b and R^c. Finally, let the price of the CM good be p_t. The central bank's budget constraint is

$$\frac{\mathbb{M}_{t+1} - \mathbb{M}_t}{p_t} + \frac{\mathbb{C}_{t+1} - \mathbb{C}_t}{p_t} + \frac{(1 + R_t^b)\mathbb{B}_t^c}{p_t} = \frac{R^c \mathbb{C}_t}{p_t} + \frac{\mathbb{B}_{t+1}^c}{p_t} + \mathcal{E}_t + \mathcal{T}_t^c. \tag{2.3}$$

The left-hand side of (2.3) represents the per-period income of the central bank, which consists of the revenue from issuing new cash and CBDC, and the revenue from redeeming the government

[5] In Appendix D.2, I consider a general functional form for the audit cost. Under certain assumptions, the main results in this paper remain unchanged.

[6] This assumption simplifies the analysis and allows me to focus on the issue of money laundering. The main mechanisms in this paper do not depend on this assumption.

Electronic copy available at: https://ssrn.com/abstract=4394190

bonds purchased in the last period. The expenses of the central bank consist of the interest payment on CBDC, the purchase of government bonds, and the cost of operating the central bank $\mathcal{E}_t$, which is taken as exogenous. After paying the expenses, the central bank transfers the rest of its income to the fiscal authority ($\mathcal{T}_t^c$).

Finally, denote the total supply of government bonds in a period as $\mathbb{B}_t$. The budget constraint of the fiscal authority is

$$\mathbb{B}_{t+1}/p_t + \tau_t + \mathcal{F}_t + \mathcal{T}_t^c = \mathcal{A}_t + (1 + R_t^b)\mathbb{B}_t/p_t, \tag{2.4}$$

where τ_t represent a lump-sum tax (or transfer if it is negative) collected from buyers and sellers in the CM, $\mathcal{F}_t$ represents the assets confiscated from sellers who launder money, and $\mathcal{A}_t$ represents the total audit costs in the DM. Then, the left-hand side of (2.4) represents the per-period income of the fiscal authority, while the right-hand side represents the expenses. Following Andolfatto and Williamson (2015) and Williamson (2016, 2019, 2022), I assume that the fiscal authority chooses the total amount of government debt ($\mathcal{D} = \frac{(1+R_t^b)\mathbb{B}_t}{p_t}$) and lets τ_t adjust passively so that (2.4) holds. The central bank determines the supply of cash and CBDC through open market operations to achieve a certain inflation target $\mu = (p_{t+1} - p_t)/p_t$.[7]

3 A Benchmark Model: No CBDC

As a benchmark, I assume the only government-issued money is cash. I also restrict my attention to stationary equilibria where all real variables remain constant.

3.1 Government Audits and Sellers' Problem

I. Sellers who engage in money laundering

I begin by considering sellers who engage in money laundering. Let R^d denote the nominal deposit rate. Suppose a buyer decides to visit market (x, p) in the next DM, where $p = (m, d)$ represents the payment in cash and deposits (in real terms), and $x = q$ represents the amount of DM good produced. To aid the description of sellers' problem, I first assume that a seller may accept either cash or bank deposits. Then, I allow sellers to accept both payment instruments.

First, suppose $d = 0$ so the seller only accepts cash. Due to inflation, accumulating one unit of cash requires $1 + \mu$ units of labor in the CM. Recall that a buyer meets a seller with probability $\min\{1, 1/n(q, p)\}$. Then, the expected surplus of a buyer visiting (q, p) is given by

$$-(1 + \mu)m + \beta \min\{1, 1/n(q, p)\}[u(q) - m] + \beta m. \tag{3.1}$$

Sellers maximize their utility by choosing q and m subject to (q, p) providing expected surplus

[7] Treating the fiscal authority and the central bank as separate entities is important for understanding the effects of CBDC on the supply of payment instruments as well as the implications for aggregate welfare.

Electronic copy available at: https://ssrn.com/abstract=4394190

equal to $\mathcal{S}$ to buyers, where $\mathcal{S}$ is the expected surplus a buyer can obtain from her best alternative. Let $\mathbb{E}(\eta)$ denote a seller's expected audit probability. Let λ represent the amount of illegal income that can be laundered per unit of legal income. Recall that money laundering generates a proportional cost $\mathcal{C}$ and a proportional payoff $\mathcal{P}$ for the seller. Let $P \equiv \mathcal{P} - \mathcal{C}$ denote the net payoff per unit of illegal income laundered. The seller solves

$$\max_{q,m} \ \min\{1, n(q,m)\}\{[1-\mathbb{E}(\eta)](1+\lambda P)m + \mathbb{E}(\eta)(1-\rho^m)(1+\lambda P)m - q\} \tag{3.2}$$

$$\text{s.t.} \ -(1+\mu)m + \beta \min\{1, 1/n(q,m)\}[u(q)-m] + \beta m = \mathcal{S}. \tag{3.3}$$

In words, (3.2) says that a seller trades a buyer with probability $\min\{1, n(q,m)\}$. After trading, if the seller is not audited (which happens with probability $[1-\mathbb{E}(\eta)]$), she receives m units of cash from the buyer plus $\lambda P m$ units of payoff from money laundering. If the seller is audited, with probability ρ^m, her money laundering activities are discovered, and her assets are confiscated. Finally, to produce q units of the DM good, the seller must supply q units of labor. It is straightforward to show that the solution is given by

$$m = \frac{\beta u(q) - \mathcal{S}}{1+\mu}, \tag{3.4}$$

where q solves

$$\beta u'(q) = \frac{1+\mu}{[1-\mathbb{E}(\eta)\rho^m](1+\lambda P)}. \tag{3.5}$$

Now, assume $m = 0$, so the seller only accepts bank deposits. Sellers' problem becomes

$$\max_{q,d} \ \min\{1, n(q,d)\}\{[1-\mathbb{E}(\eta)](1+\lambda P)d - q\} \tag{3.6}$$

$$\text{s.t.} \ -\frac{(1+\mu)d}{1+R^d} + \beta \min\{1, 1/n(q,d)\}[u(q)-d] + \beta d = \mathcal{S}. \tag{3.7}$$

Compared to problem (3.2), there are two differences: (1) if $R^d > 0$, then to buyers, the unit cost of deposits is lower than that of cash; and (2) if a seller is audited, her money laundering activities are discovered with probability one. The solution is given by

$$d = \frac{\beta u(q) - \mathcal{S}}{(1+\mu)/(1+R^d)}, \tag{3.8}$$

where q solves

$$\beta u'(q) = \frac{(1+\mu)/(1+R^d)}{[1-\mathbb{E}(\eta)](1+\lambda P)}. \tag{3.9}$$

Finally, assume that sellers can accept both cash and bank deposits. Sellers' problem is:

$$\max_{q,m,d} \min\{1, n(q,m,d)\}\{[1-\mathbb{E}(\eta)](1+\lambda P)(m+d) + \mathbb{E}(\eta)(1-\rho^m)(1+\lambda P)m - q\} \tag{3.10}$$

$$\text{s.t.} \ -(1+\mu)m - \frac{(1+\mu)d}{1+R^d} + \beta \min\{1, 1/n(q,m,d)\}[u(q)-(m+d)] + \beta(m+d) = \mathcal{S}, \tag{3.11}$$

Electronic copy available at: https://ssrn.com/abstract=4394190

It is straightforward to verify that the seller will only accept cash (i.e., $d = 0$) if

$$\frac{(1+\mu)/(1+R^d)}{[1-\mathbb{E}(\eta)](1+\lambda P)} > \frac{1+\mu}{[1-\mathbb{E}(\eta)\rho^m](1+\lambda P)}, \tag{3.12}$$

II. Sellers who do not engage in money laundering

Consider a seller who does not engage in money laundering and posts (x, p), where $x = g$ represents the DM good produced. She solves the following problem.

$$\max_{g,m,d} \ \min\{1, n(g,m,d)\}\{m + d - g\} \tag{3.13}$$

$$\text{s.t.} \ -(1+\mu)m - \frac{(1+\mu)d}{1+R^d} + \beta \min\{1, 1/n(g,m,d)\}\{u(g) - (m+d)\} + \beta(m+d) = \mathcal{S}. \tag{3.14}$$

Note that sellers who do not engage in money laundering are not affected by government audits. I assume sellers choose bank deposits if they are indifferent. Then, as long as $R^d \geq 0$, the seller will accept only bank deposits. The solution is given by

$$d = \frac{\beta u(g) - \mathcal{S}}{(1+\mu)/(1+R^d)}, \tag{3.15}$$

where g solves

$$\beta u'(g) = \frac{1+\mu}{1+R^d}. \tag{3.16}$$

Finally, in the CM, sellers choose to post either (q, m, d) or (g, m, d) to maximize their surplus.

III. Government Audits

Consider the government's choice of audit probability, η. First, note that sellers who engage in money laundering accept either cash or bank deposits. Hence, the total amount of illegal income laundered is equal to either λm or λd. Next, recall that laundered illegal income is used to fund future criminal activities, which generate a welfare cost proportional to the quantity of illegal income. Let $\tilde{N}$ denote the welfare cost generated by one unit of laundered illegal income. Define $N = \tilde{N}\lambda$. Let γ denote the share of sellers who engage in money laundering. Given the audit cost $\tilde{C}$ and its beliefs about m, d, and γ, the government chooses η to minimize the welfare cost associated with laundered illegal income.

$$\begin{aligned}\min_{\eta} \ & -\alpha\gamma \min\{1, n(q,m,d)\}\{(1-\eta)[\mathbf{1}(m>0)mN + \mathbf{1}(d>0)dN] + \eta\mathbf{1}(m>0)(1-\rho^m)mN\} \\ & - \eta\alpha\tilde{C}.\end{aligned} \tag{3.17}$$

Note that α is the total measure of sellers, and $\mathbf{1}(.)$ is an indicator function. To understand (3.17), first note that $\gamma \min\{1, n(q,m,d)\}$ represents the share of sellers who chose to engage in money laundering *and* have met buyers in the DM. Among such sellers, with probability $1-\eta$, a seller is not audited, and the welfare cost associated with laundered illegal income is equal to $-[\mathbf{1}(m>0)mN + \mathbf{1}(d>0)dN]$. If a seller is audited, the welfare cost is avoided unless the seller

Electronic copy available at: https://ssrn.com/abstract=4394190

accepted cash, and the cash income is not discovered, which happens with probability $1-\rho^m$. Finally, $\eta\alpha$ represents the total measure of sellers who are audited by the government, so $\eta\alpha\tilde{C}$ is the total audit cost. In what follows, let $C \equiv \alpha\tilde{C}$.

Now, I solve for the *equilibrium of the audit game*, which is defined as follows.
(1) Given η, DM postings (q, m, d) and (g, m, d) solve (3.10) and (3.13), respectively. Sellers choose the postings that offer the highest payoff.
(2) Given γ, (q, m, d), and (g, m, d), the government's choice of η solves (3.17).
The following proposition summarizes the equilibrium.

Proposition 1 *There exist $\underline{\rho^m}$, $\overline{\rho^m}$, $\overline{C}$, $\overline{R^d}$, and $\bar{\mu}$ such that if $\rho^m \in (\underline{\rho^m}, \overline{\rho^m})$, $C < \overline{C}$, $R^d < \overline{R^d}$, and $\mu < \bar{\mu}$, there is a unique equilibrium of the audit game where both sellers and the government play mixed strategies. Specifically,*
(1) *With probability $1-\gamma$, a seller chooses not to engage in money laundering and posts (g, m, d), where $m = 0$, $d = g$ and g solves*

$$u'(g) = \frac{1+\mu}{\beta(1+R^d)}, \tag{3.18}$$

(2) *With probability γ, a seller chooses to engage in money laundering and posts (q, m, d), where q solves*

$$u'(q) = \frac{1+\mu}{\beta[(1-\eta\rho^m)(1+\lambda P)]}, \tag{3.19}$$

$d = 0$, and m is given by

$$m = \frac{\beta u(q) - \mathcal{S}}{1+\mu}. \tag{3.20}$$

(3) *Audit probability η is such that $(1-\eta\rho^m)(1+\lambda P)m - q = 0$, and $\mathcal{S} = \beta u(g) - \frac{(1+\mu)d}{1+R^d}$.*
(4) *Finally, $\gamma = \frac{C}{\rho^m m N}$, and $n(g, m, d) = n(q, m, d) = \frac{1}{\alpha}$.*
Proof: see Appendix B.

As long as ρ^m is not too small (i.e., $\rho^m > \underline{\rho^m}$), sellers will not engage in money laundering if the government audits them with probability one. However, if sellers do not engage in money laundering, the government will have no incentive to conduct audits, in which case all sellers will strictly prefer to engage in money laundering. Hence, the unique equilibrium is a mixed strategy equilibrium, where some sellers choose to engage in money laundering, and the government audits sellers with probability strictly between zero and one. The condition that $\rho^m < \overline{\rho^m}$ ensures that sellers who engage in money laundering prefer cash. Note also that because by assumption there are more sellers than buyers, the competition between sellers drives their expected surplus to zero. Finally, if both R^d and μ are high, there may not exist an equilibrium where cash is used as payment. In such a case, the cost of using cash is so high relative to the cost of using bank deposits that sellers who engage in money laundering prefer bank deposits.

Electronic copy available at: https://ssrn.com/abstract=4394190

3.2 Bankers' Problem in the CM

Recall that the markets for bank and deposits bank loans are perfectly competitive, so bankers take the nominal deposit rate R^d and the nominal loan rate R^k as given.

Recall that bankers have limited commitment. I assume that if a banker chooses to default, he or she can abscond with a fraction θ of the collateral. Bankers' problem is given by

$$\max_{d^B, m^B, b^B, k^B} \{\beta(k^B + m^B + b^B - d^B) - e\} \tag{3.21}$$

$$\text{s.t. } \frac{(1+\mu)k^B}{1+R^k} + (1+\mu)m^B + \frac{(1+\mu)b^B}{1+R^b} = \frac{(1+\mu)d^B}{1+R^d} + e, \tag{3.22}$$

$$(k^B + m^B + b^B)(1-\theta) \geq d^B, \tag{3.23}$$

$$k^B, m^B, b^B, d^B \geq 0. \tag{3.24}$$

The amount of bank deposits (in real term) created by a banker is d^B, and e is the amount of CM good produced by the banker using the same technology as buyers. This means that to satisfy the incentive constraint (3.23), bankers may supply their own capital (i.e., "sweat equity"). A banker's holdings of cash, government bonds, and loans are m^B, b^B, and k^B, respectively.

Given R^d, the demand for deposits by a buyer who decides to visit (g, d) is $d = g$, where

$$u'(g) = \frac{1+\mu}{\beta(1+R^d)}. \tag{3.25}$$

Since a fraction $1-\gamma$ of sellers accept deposits in the DM, to clear the deposit market, R^d must be such that $d^B = (1-\gamma)d$.

Given R^k, the demand for loans by entrepreneurs is given by

$$f'(k) = \frac{1+R^k}{1+\mu}. \tag{3.26}$$

Recall that there is measure one of entrepreneurs and bankers. Hence, R^k must be such that

$$\frac{(1+\mu)k^B}{1+R^k} = k. \tag{3.27}$$

Finally, depending on the return on loans, government bonds, and cash, the bankers may hold one or more types of assets as collateral. If bankers hold both government bonds and loans, then R^k and R^b must be equal. In what follows, I focus on equilibria where constraint (3.23) binds, which happens if the real return from bank assets is less that $1/\beta$. Note also that because the loan and deposit markets are perfectly competitive, in equilibrium, expected bank profit must be zero. Now, substitute in the zero profit condition and binding constraint (3.23) into budget constraint (3.22). We have the following relationship between (R^k, R^b) and R^d.

$$\frac{1+\mu}{1+R^b} = \frac{1+\mu}{1+R^k} = \beta\theta + \frac{(1-\theta)(1+\mu)}{1+R^d}. \tag{3.28}$$

Electronic copy available at: https://ssrn.com/abstract=4394190

3.3 General Equilibrium

In this section, I solve the general equilibrium of the model economy. Recall that $\mathcal{D}$ represents the total amount of government debt (in real terms) issued by the fiscal authority, and μ is the central bank's inflation target. Given government policy $(\mathcal{D}, \mu)$, a general equilibrium is a vector $\{g, q, \gamma, \eta, k, R^d, R^b, R^k\}$ that satisfies (a) the equilibrium conditions of the audit game as described in Proposition 1; (b) the solution to bankers' problem as described in Section 3.2; and (c) budget constraints (2.3) and (2.4).

Proposition 2 *Suppose that $kf'(k)$ is increasing in k. Then, there exists a unique equilibrium.*
Proof: see Appendix B.

Next, I consider the effects of ρ^m and μ on the equilibrium.

Proposition 3 (1) *Suppose $\frac{1}{1-\theta} > 1 + \lambda$. An increase in ρ^m leads to decreases in both q and γ. In addition, g and $\frac{1+R^b}{1+\mu}$ also decrease, while k increases.*
(2) *There exists μ' such that if $\mu > \mu'$, there does not exist a monetary equilibrium where cash is accepted as payment.*
Proof: see Appendix B.

An increase in ρ^m makes money laundering easier to detect. This leads to decreases in both the goods produced by a seller who engages in money laundering, q, and the share of such sellers, γ. However, the change in ρ^m also affects the output by sellers who do not engage in money laundering through a general equilibrium effect. Specifically, because such sellers accept only bank deposits, the decrease in γ increases the demand for deposits and consequently bankers' demand for government bonds. This in turn leads to decreases in real deposit rate $\frac{1+R^b}{1+\mu}$ and real bond rate. As a result, sellers who accept bank deposits produce less for buyers. However, a lower government bond rate also means a lower loan rate (since loan rate is always equal to bond rate in equilibrium), which increases capital investment.

The second part of Proposition 3 says that sufficiently high inflation can drive cash out of the model economy. When inflation rises, the costs of holding cash will increase, and so will the costs of holding bank deposits if the nominal deposit rate, R^d, remains unchanged. However, in such a scenario, bankers have the incentive to increase R^d and attract bank deposits, which fund the investments in government bonds and entrepreneur loans. Consequently, when μ is sufficiently high, the cost of using cash becomes too large relative to the cost of using bank deposits so that even sellers who launder money will prefer bank deposits over cash.

However, it is worth noting that such a non-monetary equilibrium is merely an artifact of the assumption that only cash and bank deposits are available as payment instruments. It is straightforward to add foreign currencies and cryptocurrencies such as bitcoin to the benchmark model. If the inflation of the domestic cash is sufficiently high, sellers who engage in money

Electronic copy available at: https://ssrn.com/abstract=4394190

laundering will switch to such alternative payment instruments provided that they offer relatively low inflation and comparatively low probability of detection in the event of government audits. I discuss this scenario further in Section 7. In what follows, the monetary equilibrium where cash is used by sellers who launder money will serve as the benchmark.

4 Central Bank Digital Currency: Equilibrium

I define central bank digital currency (CBDC) to be another type of payment instrument (in addition to cash) that is issued by the central bank. It is stored in accounts managed by the central bank, and it can be held by all agents including sellers, buyers, bankers, and entrepreneurs. However, I will show later that bankers in general do not hold CBDC, because it offers a lower return compared to government bonds and bank loans. Throughout this section, I assume that the central bank does not withdraw cash from circulation. This assumption is in line with the Federal Reserve's current policy, which treats CBDC as a means to expand payment options rather than replace cash.[8] Although cash continues to be available from the central bank, I will show that depending on the characteristics of CBDC, cash may not be used in equilibrium, as CBDC may be a superior payment instrument.

It is also important to be clear about how CBDC is introduced into the economy. Recall that in the benchmark, the central bank can increase or decrease the supply of cash through open market purchases or sales of government bonds. The central bank's balance sheet is

Assets of the central bank	**Liabilities of the central bank**
Government bonds	Cash

Similar to cash, the central bank introduces CBDC by using it to purchase government bonds from the bond market. The central bank's balance sheet after the introduction of CBDC is

Assets of the central bank	**Liabilities of the central bank**
Government bonds	Cash
	Central bank digital currency

Now, I discuss the characteristics of CBDC. I assume that CBDC can be different from cash and bank deposits in two dimensions: the interest rate on CBDC (R^c) and the amount of anonymity CBDC offers. In the context of this paper, the degree of anonymity of CBDC determines the probability of a seller's money laundering activities being discovered after an audit (denoted by ρ^c) when CBDC is accepted as payment. Offering anonymity with CBDC is technologically feasible given the currently available cryptographic technologies, and public trust in such anonymity designs can be achieved through third-party reviews of CBDC architecture

[8] See `https://www.federalreserve.gov/cbdc-faqs.htm`.

Electronic copy available at: https://ssrn.com/abstract=4394190

and operations (Darbha and Arora, 2020).[9] Based on these findings, I assume that in addition to R^c, ρ^c is also determined by the central bank. I consider scenarios where $(R^c, \rho^c) \in \mathbb{R}_+ \times [\rho^m, 1]$. Recall that ρ^m is the probability of money laundering activities being discovered if a seller's assets are in cash. Since CBDC competes with cash, if $\rho^c \geq \rho^m$, then it must be that $R^c \geq 0$, because otherwise, sellers will strictly prefer cash over CBDC.

There are four types of equilibria that exist depending on R^c and ρ^c. A full characterization of all equilibria can be found in Appendix A. These equilibria differ in the payment instrument(s) accepted by sellers who engage in money laundering and sellers who do not.

Equilibrium Types	Accepted by Sellers **who launder money**	Accepted by Sellers **who do not launder money**
Type-1	Cash, or cash and CBDC	Bank deposits
Type-2	CBDC	Bank deposits
Type-3	CBDC	CBDC and bank deposits
Type-4	Cash	CBDC and bank deposits

Table 1: Equilibrium Types

The figure below shows how equilibrium types depend on the central bank's choices R^c and ρ^c. The origin represents $(R^c, \rho^c) = (0, \rho^m)$, and R^{bench} represents the deposit rate in the benchmark equilibrium (see Section 3).

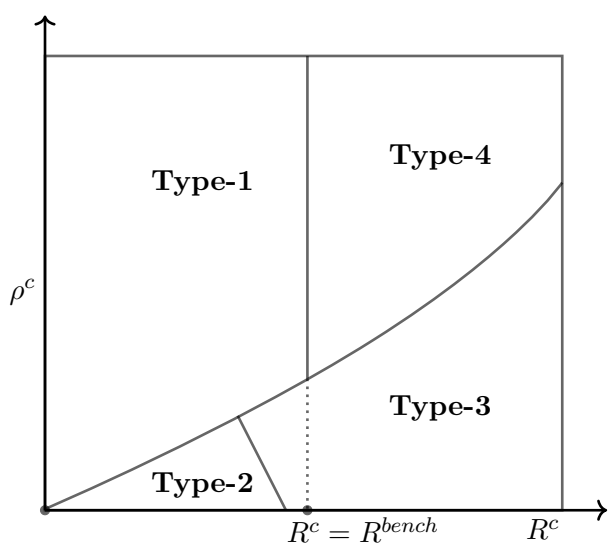


Figure 1: Equilibrium Types

In type-1 equilibria, sellers are indifferent between cash and CBDC, so equilibrium outcomes are similar to the benchmark. In what follows, I focus on the other equilibrium types and show how R^c and ρ^c affect key equilibrium outcomes. I discuss welfare implications in Section 5.

[9] A report by European System of Central Banks (ESCB (2019)) also shows how CBDC can offer anonymity through "anonymity vouchers".

Electronic copy available at: https://ssrn.com/abstract=4394190

4.1 Type-2 Equilibrium

In a type-2 equilibrium, ρ^c and R^c are such that sellers who engage in money laundering prefer to use CBDC, but because R^c is lower than the deposit rate, sellers who do not engage in money laundering prefer to accept bank deposits. The following proposition describes the effects of R^c and ρ^c on the equilibrium.

Proposition 4 *Increasing R^c or ρ^c while holding μ constant lowers γ and q. In addition, g, R^d, and R^b will decrease, while k will increase.*
Proof: see Appendix B.

For a given audit probability η, a higher R^c increases the legal revenue of sellers who engage in money laundering and allows them to launder more illegal income, which increases the welfare cost of money laundering. In such a case, the government has the incentive to increase audit probability. This lowers the share of sellers who launder money and decreases their output. As more sellers move away from money laundering, the demand for deposits increases, which drives down the government bond rate and deposit rate. As a result, the output by sellers who do not engage in money laundering also decreases, although the decreases in interest rates lead to more bank lending. Finally, a higher ρ^c makes money laundering easier to detect, so the government has more incentive to audit sellers, which has the same effect as an increase in R^c: the share of sellers who launder money decreases, while the demand for deposits increases.

Figure 2 illustrates the mechanism. Note that an increase in R^c or ρ^c does not directly affect the output of sellers who do not engage in money laundering, since they only accept bank deposits. However, in a general equilibrium, the decrease in money laundering affects the demand for bank deposits and the deposit rate, which in turn affects g.

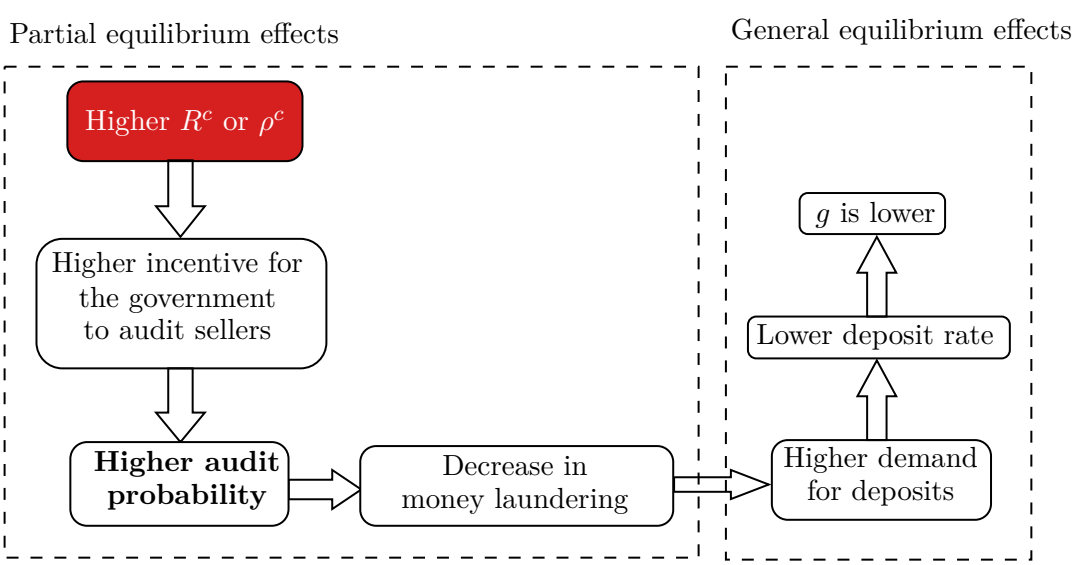


Figure 2: Effects of Increasing R^c or ρ^c in a Type-2 Equilibrium

Electronic copy available at: https://ssrn.com/abstract=4394190

4.2 Type-3 Equilibrium

Before I discuss type-3 equilibria, it should be made clear that while CBDC may completely replace cash (a type-2 equilibrium), it will not completely replace bank deposits. This is because if deposits are not accepted in equilibrium, bankers will have to fund the loans to entrepreneurs through working in the CM. In such a case, the real loan rate $\frac{1+R^k}{1+\mu}$ and the real deposit rate will both be equal to $\frac{1}{\beta}$. This means that unless the real interest rate on CBDC is greater or equal to $\frac{1}{\beta}$, sellers will prefer to use bank deposits. However, the demand for CBDC will be infinite if the real interest rate on CBDC is greater than $\frac{1}{\beta}$. Hence, in any equilibrium, sellers who do not engage in money laundering will either strictly prefer bank deposits or be indifferent between CBDC and bank deposits, but they will not strictly prefer CBDC.

Now, I discuss the effects of R^c and ρ^c on the equilibrium. Among the sellers who do not engage in money laundering, let ϵ be the faction of sellers who accept bank deposits.

Proposition 5 (1) *Increasing R^c while holding μ and ρ^c constant decreases γ and but increases g and q. In addition, ϵ and k will decrease.* (2) *Increasing ρ^c while holding μ and R^c constant has no effect on g, q, and k. However, γ and ϵ will decrease.*
Proof: see Appendix B.

A higher R^c increases the income of all sellers including those who engage in money laundering, which allows them to launder more illegal income. This gives the government more incentive to audit sellers unless fewer sellers choose to launder money. Hence, in equilibrium, γ decreases while g and q increase. A higher ρ^c makes money laundering activities easier to detect and lowers sellers' incentive to engage in money laundering. Because fewer sellers choose to launder money, the government has less incentive to audit sellers. Because of the decrease in audit probability, q remains constant despite the increase in ρ^c.

To understand why ϵ decreases with R^c and ρ^c, first recall that CBDC is created by the central bank through purchasing government bonds with CBDC, so one unit of government bonds can be used to create one unit of CBDC. Because bankers have limited commitment and can abscond with a fraction θ of assets, they can only create $1-\theta$ units of bank deposits with each unit of government bonds. This means that the central bank is more efficient at creating payment assets compared to bankers. In equilibrium, a higher R^c or ρ^c reduces money laundering and increases the demand for bank deposits and CBDC. To satisfy the demand, the more efficient creator of payment assets, the central bank, increases the supply of CBDC by purchasing more government bonds from the bond market, while bankers decrease their holdings of government bonds and reduce the supply of deposits. As a result, in equilibrium, a larger share of sellers who do not engage in money laundering choose to accept CBDC.

It should be noted that in both type-2 and type-3 equilibria, a higher R^c decreases γ. However, in a type-3 equilibrium, R^d, g, and q are increasing in R^c, but in a type-2 equilibrium,

Electronic copy available at: https://ssrn.com/abstract=4394190

they are decreasing in R^c. This is because, in a type-2 equilibrium, CBDC is not accepted by sellers who do not engage in money laundering. A decrease in γ leads to a higher demand for deposits, which decreases the deposit rate and lowers sellers' output. In a type-3 equilibrium, a decrease in γ also increases the demand for deposits, but since the interest rate on CBDC is high, some sellers who do not engage in money laundering are willing to substitute CBDC for bank deposits. Since the central bank is more efficient at creating payment assets, the total supply of payment assets (CBDC plus bank deposits) increases. This allows sellers who do not engage in money laundering to produce more goods in equilibrium.

4.3 Type-4 Equilibrium

In a type-4 equilibrium, CBDC is not accepted by sellers who engage in money laundering, so how much anonymity CBDC offers does not affect equilibrium outcomes. Now, consider the effects of R^c. Among the sellers who do not engage in money laundering, let ϵ be the faction of sellers who accept bank deposits.

Proposition 6 *Increasing R^c while holding μ constant decreases γ but increases g and q. In addition, ϵ and k will decrease.*
Proof: see Appendix B.

Increasing R^c has similar effects in type-3 and type-4 equilibria. Specifically, a higher R^c increases the income and output of sellers who do not engage in money laundering and lowers the incentive for sellers to launder money. The decrease in γ lowers the government's incentive to audit but increases the demand for bank deposits and CBDC. Since the central bank is more efficient in creating payment assets, ϵ decreases in equilibrium.

5 Aggregate Welfare and Central Bank Net Revenue

In the first part of this section, I discuss the welfare effects of introducing a central bank digital currency (CBDC). Aggregate welfare is measured using an equal-weighted sum of all agents' utility. Let X denote the total consumption in the CM and L denote the total labor supplied in the CM. Define $\mathcal{N}$ to be the total welfare cost generated by laundered illegal income. Aggregate welfare is then given by

$$\mathcal{W} = (1-\gamma)[u(g)-g] + \gamma[u(q)-q] + X - L - \mathcal{N}. \tag{5.1}$$

Recall that g and q represent the output of a seller who does not engage in money laundering and the output of a seller who does, respectively, and γ denotes the share of sellers who engage in money laundering.

In the second part of this section, I discuss the effect of introducing CBDC on the central

Electronic copy available at: https://ssrn.com/abstract=4394190

bank's net revenue. The central bank in the model, similar to its counterparts in reality (e.g., the Federal Reserve), earns revenue through the interest payments on its assets (e.g., government bonds). The interest expenses of the central bank are the interest payments on its liabilities (CBDC and/or cash). To ensure a central bank's independence from the fiscal authority, it may be necessary for the central bank to cover its expenses rather than rely on transfers from the fiscal authority.[10] It is therefore of interest to understand how introducing CBDC, especially interest-bearing CBDC, affects the central bank's net revenue.

5.1 Aggregate Welfare

Recall that η is the audit probability, and ηC is the total audit cost. Recall also that k is the amount of CM good invested in each entrepreneur's project. Then, $\beta f(k) - k$ represents the net output from entrepreneurs' projects. Hence, the net consumption of the CM good (i.e., $X - L$) must be equal to

$$X - L = \beta f(k) - k - \eta C. \tag{5.2}$$

Now, consider the total welfare cost generated by laundered illegal income. Let $\mathcal{A}$ denote the income received by a seller who engage in money laundering. Without government audits, the welfare cost is equal to $\gamma \mathcal{A} N$. However, a fraction $\eta\rho$ of sellers who engage in money laundering is discovered by the government, where $\rho = \rho^m$ if cash is used and $\rho = \rho^c$ if CBDC is used. Hence, we have

$$\mathcal{N} = (1 - \eta\rho)\gamma \mathcal{A} N. \tag{5.3}$$

Aggregate welfare can then be divided into three components.

$$\mathcal{W} = \underbrace{(1-\gamma)[u(g) - g] + \gamma[u(q) - q]}_{\text{Total surplus in the DM}} + \underbrace{\beta f(k) - k}_{\text{Entrepreneur net output}} - \underbrace{[\eta C + (1 - \eta\rho)\gamma \mathcal{A} N]}_{\text{Total cost from money laundering}}. \tag{5.4}$$

The welfare effect of introducing CBDC is in general ambiguous. As shown in Section 4, g and q always move in the opposite direction of k when R^c or ρ^c changes. To derive clear welfare implications, I restrict my attention to some subsets of CBDC configurations.

First, assume $R^c > R^{bench}$, where R^{bench} represents the deposit rate in the benchmark equilibrium. Recall that in this case, only type-3 and type-4 equilibria exist (see Figure 1). Now, define $\rho(R^c)$ to be the highest ρ^c such that given (R^c, ρ^c), the equilibrium is type-3. Specifically, $\rho(R^c)$ solves

$$\frac{1+\mu}{[1 - \rho(R^c)\eta](1 + R^c)} = \frac{1+\mu}{1 - \rho^m \eta}, \tag{5.5}$$

[10] For example, when discussing independence, the Federal Reserve emphasizes that it "does not receive funding through the congressional budgetary process". (`https://www.federalreserve.gov/fqs/about_12799.htm`)

Electronic copy available at: https://ssrn.com/abstract=4394190

where η is the equilibrium audit probability. In words, $\rho(R^c)$ is the highest value of ρ^c such that sellers who engage in money laundering prefer CBDC over cash.

Proposition 7 *For any $R^c > R^{bench}$, $(R^c, \rho(R^c))$ offers the highest aggregate welfare.*
Proof: see Appendix B.

In type-3 and type-4 equilibria, $R^c = R^d$ (bank deposit rate). From Section 3 and 4, we know that g, q, and k only depend on R^d. This means that for any given $R^c > R^{bench}$, aggregate welfare varies only because of ρ^c. In the proof of Proposition 7, I show that when $\rho^c = \rho(R^c)$, the share of sellers who engage in money laundering, γ, reaches its minimum, and so does the total cost from money laundering. In other words, for any given $R^c > R^{bench}$, aggregate welfare is maximized when CBDC offers the least amount of anonymity such that sellers who engage in money laundering accept CBDC instead of cash.

Second, assume $R^c \leq R^{bench}$. In this scenario, the effect of ρ^c on aggregate welfare is in general ambiguous even if we fix R^c. This is because, in a type-2 equilibrium, an increase in ρ^c lowers the bank deposit rate, which has opposite effects on (g, q) and k. Nevertheless, if the question we wish to answer is whether introducing CBDC *can* increase aggregate welfare, then we may restrict our attention to R^c that is equal to R^{bench}. In such cases, k is equal to its counterpart in the benchmark equilibrium. I refer to this type of CBDC as "**deposit-like**" CBDC. Formally, a deposit-like CBDC satisfies $R^c = R^{bench}$ and $\rho^c = \rho(R^c)$. Recall that this means CBDC offers the least amount of anonymity such that sellers who engage in money laundering prefer CBDC over cash. I compare such a CBDC with "**cash-like**" CBDC. Formally, a cash-like CBDC satisfies
(1) $0 \leq R^c \leq R^{c\prime}$ and $\rho^m \leq \rho^c \leq \rho^{c\prime}$ for some $0 < R^{c\prime} < R^{bench}$ and $\rho^{c\prime} > \rho^m$, and
(2) $\frac{1+\mu}{[1-\rho(R^c)\eta](1+R^c)} > \frac{1+\mu}{1-\rho^m\eta}$, where η is the equilibrium audit probability.
In other words, similar to cash, CBDC offers a low return but high anonymity. As a result, sellers who do not engage in money laundering accept deposits. Nevertheless, (R^c, ρ^c) are such that sellers who engage in money laundering prefer CBDC over cash. The following figure shows the locations of deposit-like and cash-like CBDC in the R^c-ρ^c equilibrium space.

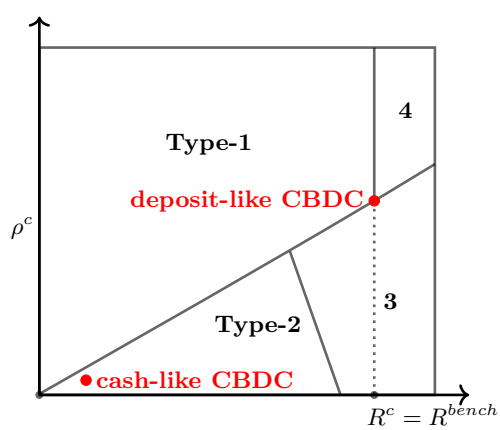


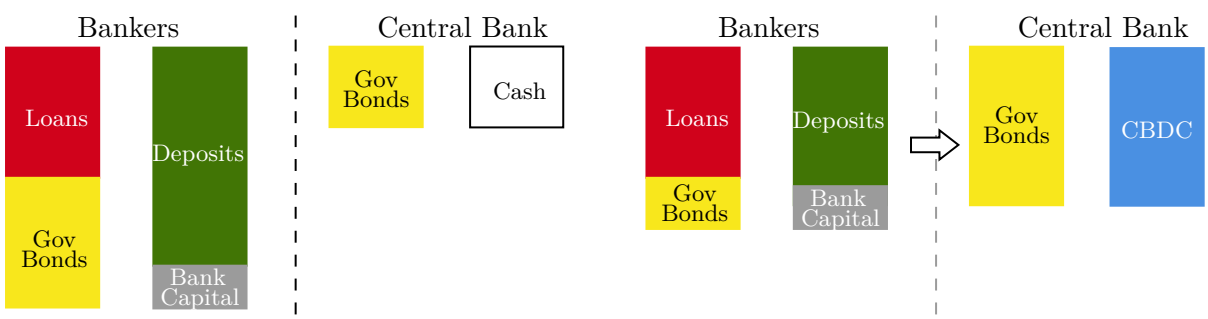
Figure 3: deposit-like and cash-like CBDC

Electronic copy available at: https://ssrn.com/abstract=4394190

Proposition 8 *Then, introducing deposit-like CBDC increases aggregate welfare. Furthermore, there exist $R^{c\prime} > 0$ and $\rho^{c\prime} > \rho^m$ such that aggregate welfare is higher with deposit-like CBDC than with cash-like CBDC.*
Proof: see Appendix B.

To see why deposit-like CBDC provides higher aggregate welfare, note first that because money laundering generates a welfare cost, ceteris paribus, a decrease in the share of sellers who engage in money laundering, γ, increases aggregate welfare. While both types of CBDC can reduce γ, if CBDC is cash-like, sellers who do not engage in money laundering accept only bank deposits, and therefore a decrease in γ increases the demand for deposits. The resulted *shortage* of bank deposits lowers the deposit rate and lowers the output from sellers who do not engage in money laundering. To avoid this problem, the design of CBDC must achieve two objectives. First, introducing CBDC must increase the supply of payment assets in the economy. Second, the interest rate on CBDC must be sufficiently high for sellers who do not engage in money laundering to accept CBDC. Note that achieving only one of the two objectives is not sufficient to avoid the shortage of bank deposits: the shortage will remain either if CBDC is not accepted by sellers who do not engage in money laundering (e.g., cash-like CBDC), or if introducing CBDC does not increase the total supply of payment assets.

Achieving the second objective requires the interest rate on CBDC to be sufficiently high, but how does introducing CBDC increase the supply of payment assets? Recall that bankers have limited commitment so they can abscond with a portion θ of their assets. By assumption, the central bank is immune from the limited commitment problem. This means that bankers must hold more assets than their liabilities so that they do not have the incentive to default. In comparison, the central bank holds the same quantity of assets (government bonds) and liabilities (cash and/or CBDC). See Figure 4(a) for an illustration.



(a) Before Introducing deposit-like CBDC (b) After Introducing deposit-like CBDC

Figure 4: The Balance Sheet Effects of Introducing deposit-like CBDC

Recall that the central bank introduces CBDC by purchasing government bonds with CBDC. Since there are fewer government bonds available, bankers issue fewer deposits. (See Figure 4(b).) Next, note that with each unit of government bonds, the central bank can create one unit of CBDC, while bankers can only create $1 - \theta$ units of deposits. As a result, the total supply

Electronic copy available at: https://ssrn.com/abstract=4394190

of bank deposits and deposit-like CBDC is larger compared to the supply of deposits in the benchmark. In other words, the introduction of CBDC increases the supply of payment assets. This result is reminiscent of the findings in Williamson (2022), who shows that the main benefit of introducing CBDC comes from more efficient use of the aggregate stock of collateral assets.

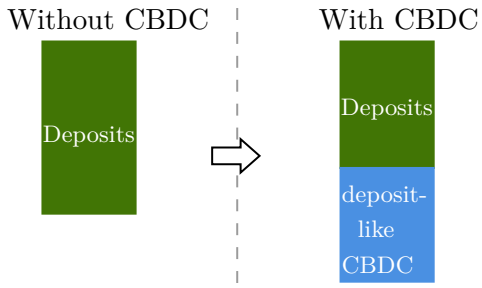


Figure 5: Supply of Bank Deposits and deposit-like CBDC

Regardless of whether CBDC is cash-like or deposit-like, introducing CBDC will decrease the share of sellers who launder money and increase the demand for payment assets. However, with deposit-like CBDC, the increase in the demand for payment assets is satisfied by the increase in the supply of payment assets. Without this effect, the increase in demand for deposits will drive down the deposit rate and decrease the output of sellers who accept bank deposits.

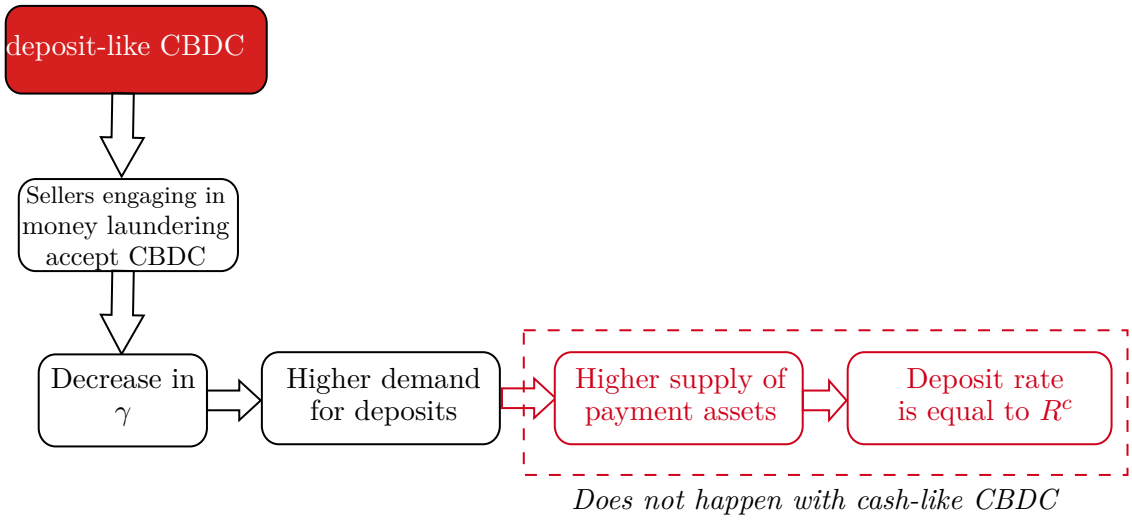


Figure 6: Effects of Introducing deposit-like CBDC

It should be noted that loans to entrepreneurs depend only on R^k, the loan rate, so if $R^c = R^{bench}$, then R^k and bank loans are unchanged following the introduction of deposit-like CBDC. It is indeed true that some bank deposits are crowded out by CBDC. However, they are part of the deposits that were originally backed by government bonds rather than bank loans (see Figure 4(b)). Nevertheless, if $R^c > R^{bench}$, introducing CBDC will raise the loan rate. In such a case, the central bank faces a trade-off: a higher R^c facilitates the exchange in the DM, but it also reduces bank loans. The same trade-off can also be found in Keister and Sanches (2023). The difference is that in Keister and Sanches (2023), CBDC can only differ from bank

Electronic copy available at: https://ssrn.com/abstract=4394190

deposits in interest rates, so for a deposit-like CBDC to have real effects, its interest rate must be different from the deposit rate in the benchmark. In this paper, CBDC may also offer a different level of anonymity, ρ^c. Proposition 8 shows that when ρ^c is chosen optimally, CBDC can increase aggregate welfare even if it has the same return as deposits in the benchmark.

5.2 Central Bank Net Revenue

The goal of this section is to understand the impact of an interest-bearing CBDC on the central bank's net revenue. The central bank's budget constraint is given by

$$\frac{\mathbb{M}_{t+1} - \mathbb{M}_t}{p_t} + \frac{\mathbb{C}_{t+1} - \mathbb{C}_t}{p_t} + \frac{(1 + R^b)\mathbb{B}_t^c}{p_t} = \frac{R^c \mathbb{C}_t}{p_t} + \frac{\mathbb{B}_{t+1}^c}{p_t} + \mathcal{E}_t + \mathcal{T}_t^c. \tag{5.6}$$

Recall that $\mathcal{E}_t$ is the central bank's operating expenses, which I take as given. Since the central bank uses the cash and CBDC it creates to purchase government bonds, we have $\mathbb{M}_t + \mathbb{C}_t = \mathbb{B}_t^c$ for all t. Now, define $C = \frac{\mathbb{C}_t}{p_t}$ and $\tilde{m} = \frac{\mathbb{M}_t}{p_t}$. The central bank's net revenue can be written as

$$\Pi = (R^b - R^c)C + R^b \tilde{m} - \mathcal{E}. \tag{5.7}$$

Note that $\tilde{m} = 0$ in type-2 and type-3 equilibria, but $\tilde{m} > 0$ in type-4 equilibria. Note also that Π is equal to the transfer to the fiscal authority, $\mathcal{T}^c$.

The effect of introducing CBDC on central bank net revenue depends on the interest rate spread $R^b - R^c$ and the demand for central bank liabilities, C and $\tilde{m}$. In a type-2 equilibrium, R^b is smaller compared to the benchmark (see Proposition 4). The demand for CBDC is also lower compared to the demand for cash in the benchmark, because CBDC is only accepted by sellers who engage in money laundering, and fewer such sellers exist in equilibrium. Hence, the central bank's net revenue is lower. In a type-3 equilibrium, the interest rate spread $R^b - R^c$ is smaller than in a type-2 equilibrium because R^c is higher. However, the demand for CBDC is also higher since CBDC is accepted by both sellers who engage in money laundering and (some) sellers who do not engage in money laundering. Consequently, the effect is in general ambiguous.

If the decrease in the central bank's net revenue is significant, the central bank may not be able to cover its own expense and may require transfers from the fiscal authority (i.e., $\Pi <$). Relying on the fiscal authority's transfers may undermine the independence of the central bank.

6 Quantitative Analysis

To gain a better understanding of the magnitude of CBDC's impact on equilibrium outcomes, I calibrate the benchmark model in Section 3 to the U.S. economy during 2016-2019. I then introduce CBDC to the economy and show how the central bank's choice of anonymity (ρ^c) and interest rate (R^c) affect money laundering, aggregate welfare, and central bank revenue.

Electronic copy available at: https://ssrn.com/abstract=4394190

6.1 Calibration

I assume that the utility function takes the form of $u(q) = \frac{q^{1-\sigma}}{1-\sigma}$, and the production function is $f(k) = k^{\upsilon}$ where $\upsilon \in (0, 1)$. There are 11 parameters to calibrate: σ, υ, β (discount factor), μ (inflation rate), ρ^m (probability that cash income is observed after an audit), C (audit cost), λ (money laundered per unit of legal income), P (unit payoff from money laundering), $\tilde{N}$ (welfare cost per unit of laundered money), θ (the fraction of assets bankers can abscond with), and $\mathcal{D}$ (total government liabilities). I also generalize the measure of entrepreneurs to S and calibrate it to data, rather than restricting it to one as in the theoretical analysis.[11]

The calibrated parameters and their targets are summarized in the following table. All targets are averages of the period 2016-2019 unless otherwise stated. I pick the period 2016-19 because the deposit rate and other interest rates were consistently higher than in the first half of the 2010s due to the Federal Reserve's policy on interest on reserves.

Param.	Values	Calibration targets	Sources
β	0.960	Standard in the literature	
σ	0.306	Money demand elasticity (1959-2007)	Lucas and Nicolini (2015); FRED
υ	0.779	Loan demand elasticity (1976-2007)	Drechsler et al. (2017); FRED
μ	1.82%	Avg. inflation (CPI): 1.82%	FRED
C	0.052	Enforcement cost per audit to GDP per capita: 0.073	Internal Revenue Service (2020)
ρ^m	0.538	ML to GDP (2014): 1.4%	Ferwerda et al. (2020)
λ	0.910	ML to domestic currency in circulation (2014): 0.476	Judson (2017), FRED
P	0.1	Est. cost to launder money (2007): 10%	Masciandaro et al. (2007)
$\tilde{N}$	3.5	Est. return from illegal investment (2007): 350%	Barone and Masciandaro (2019)
θ	0.860	Loan rate (1st percentile): 5.25%	Drechsler et al. (2017)
$\mathcal{D}$	5.613	Avg. rate on transactional deposits: 0.52%	Drechsler et al. (2017)
S	0.766	Avg. comm. & ind. loans-to-GDP ratio: 0.108	FRED

*ML stands for money laundering

Table 2: Calibration

To calibrate σ, I follow Lucas (2000) and Lagos and Wright (2005) and target the empirical money demand curve. Specifically, I use the new M1 series from Lucas and Nicolini (2015) and Moody's AAA corporate bond yield from FRED to estimate the interest rate elasticity of M1-to-nominal GDP ratio, which I then match by varying σ. Similarly, υ is chosen to match the interest rate elasticity of bank loan-to-nominal GDP ratio, which is estimated using the call reports data from Drechsler et al. (2017). I do not include the period after 2007 because after the financial crisis, M1 might have been held for non-transactional purposes (Chiu et al., 2022), while the model in this paper only focuses on transactional demand.

To calibrate audit cost C, I target the Internal Revenue Service (IRS)'s average "enforcement" expenses between 2016 and 2019, which consist of costs incurred by audit-related investi-

[11] Setting the measure of entrepreneurs to one does not affect the theoretical results in this paper. However, because household surplus and entrepreneur surplus tend not to move in the same direction following the introduction of CBDC, it is important for welfare analysis to generalize the measure of entrepreneurs.

Electronic copy available at: https://ssrn.com/abstract=4394190

gations, examinations, and collections.[12] To determine ρ^m, I target the ratio between the total amount of money laundered and GDP in the data. Ferwerda et al. (2020) estimate the ratio to be 1.4% for the U.S. in 2014. Similarly, I set λ to match the ratio between the total amount of money laundered and currency in circulation *within* the U.S. To calculate the latter, I use the results in Judson (2017), who estimates that about 40% of U.S. currency was held within the country in 2014. Finally, P and $\tilde{N}$ are set to match (1) the estimated cost (from the perspective of criminals) to launder money (Masciandaro et al., 2007) and (2) the estimated illegal return from illegal investment (Barone and Masciandaro, 2019). As pointed out by the authors, these estimates vary greatly depending on the types of criminal activities. In Appendix C, I include the quantitative results when P and $\tilde{N}$ are set to 0.2 and 7, respectively.

Finally, recall that in the model, the deposit rate adjusts so that the demand for government liabilities is equal to the supply. This allows me to identify $\mathcal{D}$ by targeting the deposit rate in data. Recall also that the relationship between the loan rate and the deposit rate is

$$\frac{1+\mu}{1+R^k} = \beta\theta + \frac{(1-\theta)(1+\mu)}{1+R^d}, \tag{6.1}$$

where R^k is the loan rate and R^d is the deposit rate. This allows me to identify θ by targeting the loan rate in data. Since there is no loan risk in the model, I follow Chiu et al. (2022) to define the rate for "safe" loans to be the first percentile of the loan rate distribution in the data.

6.2 Effects of Introducing CBDC

Figure 7 shows how equilibrium types depend on ρ^c and R^c. The boundaries between equilibrium types are depicted by solid lines. Note that type-2 equilibria only exist in the shaded area.

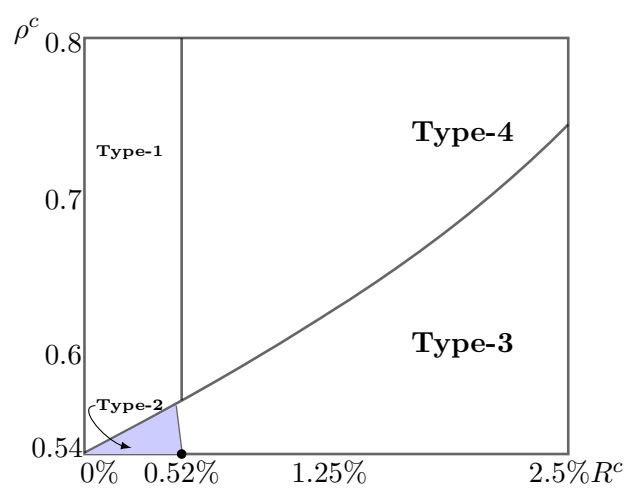


Figure 7: Equilibrium Types

[12] The purposes of the IRS' audits are to enforce both tax laws and money laundering laws. In 2019, investigations involving illegal income (such as narcotics-related income) constitute about 60% of all criminal investigations conducted by the IRS (Internal Revenue Service, 2020).

Electronic copy available at: https://ssrn.com/abstract=4394190

Recall that in the calibrated benchmark model, $\rho^m = 0.538$ and the deposit rate $R^{bench} = 0.52\%$. As predicted by the theory, only type-3 and type-4 equilibria exist when $R^c > R^{bench} = 0.52\%$.

Figure 8 shows a CBDC that offers high interest rate and low anonymity has significant impact on reducing money laundering. Figure 9 shows the effects on aggregate welfare. High-interest and low-anonymity CBDC increase welfare through two channels. First, the increase in ρ^c lowers the total cost from money laundering. Second, the increase in interest rate lowers the cost of holding CBDC and promotes trade in the DM. Recall that higher interest rate on CBDC also increases all interest rates in the economy, including the loan rate. However, in the calibrated economy, the increase in DM surplus offsets the decrease in the demand for loans, so aggregate welfare is higher. As predicted by theory, among CBDC that offer high interest rate, welfare is maximized when a CBDC offers the least amount of anonymity such that sellers who engage in money laundering prefer CBDC over cash (i.e., when (R^c, ρ^c) is on the boundary between type-3 and type-4 equilibria).

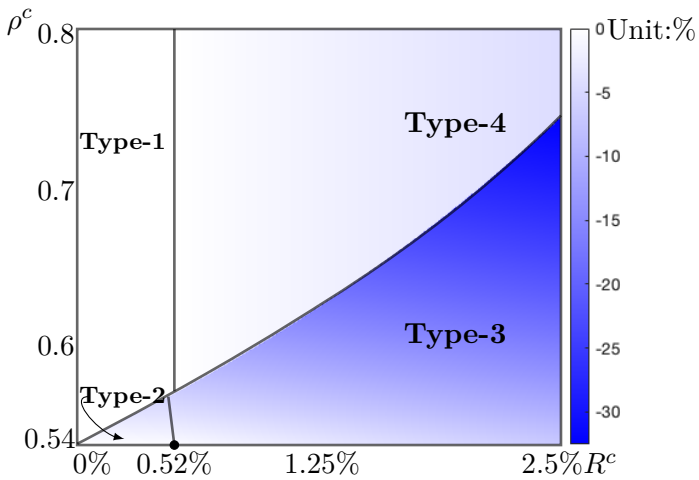


Figure 8: Percentage Change in Money Laundering (γ) (relative to the benchmark)

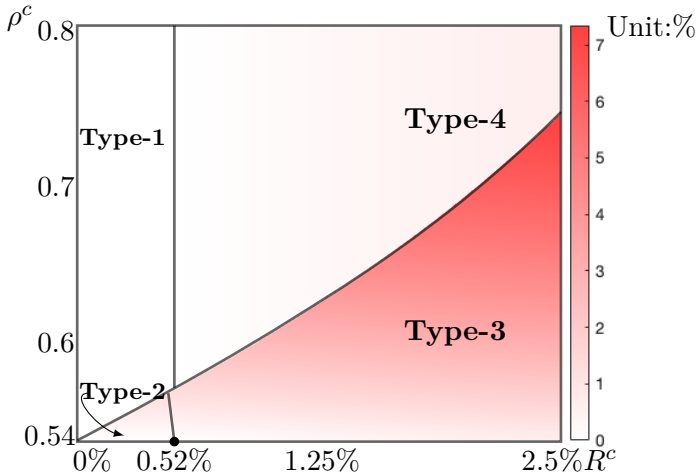


Figure 9: Percentage Change in Aggregate Welfare (relative to the benchmark)

Electronic copy available at: https://ssrn.com/abstract=4394190

In Figure 10, I show the effects of CBDC on the central bank's net revenue. Introducing CBDC with a positive interest rate reduces the central bank's seigniorage revenue per unit of payment assets, since cash pays no interest. However, a higher interest rate on CBDC also promotes trade in the DM, thereby increasing the demand for CBDC. When the interest rate on CBDC is sufficiently high, the second effect dominates, so the central bank's net revenue increases. Note that given R^c, the central bank's net revenue is higher in type-4 equilibria than in type-3 equilibria. This is because, in a type-4 equilibrium, sellers who engage in money laundering use cash, and CBDC only represents a portion of the central bank's liabilities. Since the central bank does not pay interest on cash, the net revenue is higher in this scenario.

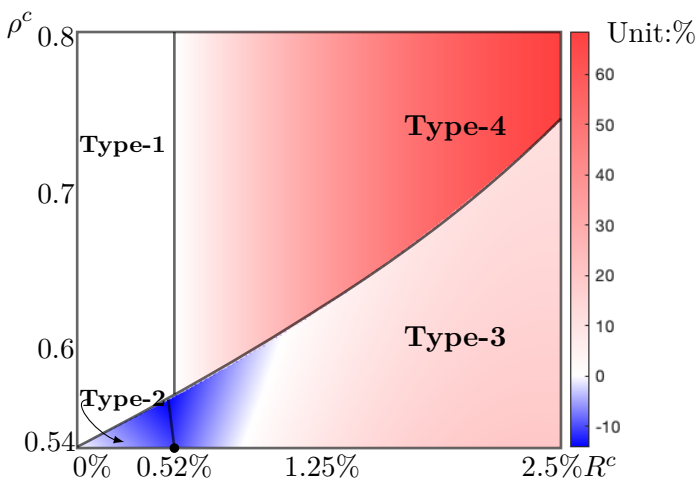


Figure 10: Percentage Change in Central Bank Net Revenue (relative to the benchmark)

7 Discussion: Non-monetary Equilibrium

The results so far are based on the assumption that the benchmark equilibrium is a monetary equilibrium where cash is accepted as payment. As shown in Proposition 3, if inflation is sufficiently high, the cost of accepting cash will be so large relative to the cost of accepting bank deposits that even sellers who engage in money laundering will prefer to accept deposits. I refer to such a scenario as a non-monetary equilibrium. It is worth mentioning that money laundering still happens in this type of equilibrium. This is because if sellers do not engage in money laundering, the government will have no incentive to conduct audits, in which case sellers will strictly prefer to launder money. However, if the government conduct audits with probability one, sellers will not engage in money laundering, since their income will be confiscated with certainty. Hence, the unique equilibrium is a still mixed strategy equilibrium, where some sellers choose to engage in money laundering, and the government audits sellers with probability strictly between zero and one.

Suppose the central bank, instead of issuing CBDC, sets inflation to be sufficiently high

Electronic copy available at: https://ssrn.com/abstract=4394190

so that the equilibrium is non-monetary. Will such a policy be welfare-improving? Recall that aggregate welfare is defined by the following:

$$\mathcal{W} = \underbrace{(1-\gamma)[u(g)-g]+\gamma[u(q)-q]}_{\text{Total surplus in the DM}} + \underbrace{\beta f(k)-k}_{\text{Entrepreneur net output}} - \underbrace{[\eta C+(1-\eta\rho)\gamma\mathcal{A}N]}_{\text{Total cost from money laundering}}. \tag{7.1}$$

Using Proposition 3, we can conclude that compared to the monetary benchmark in Section 3, the share of sellers who engage in money laundering, γ, will decrease. However, as sellers switch away from money laundering, the demand for bank deposits will increase, which will drive down the real deposit rate and lead to decreases in both the output by sellers who launder money (q) and the output by sellers who do not (g). Nevertheless, because loan rate will be lower, capital investment (k) will increase. We also know from Section 5.1 that because deposits are easier to detect, the total cost from money laundering will decrease.

While it is clear how each component of aggregate welfare will change, the overall effects of a non-monetary equilibrium are in general ambiguous, because not all components move in the same direction. In what follows, I use the parameters calibrated in Section 6 to solve for the non-monetary equilibrium and compare it with the monetary benchmark as well as equilibria with different configurations of CBDC. Specifically, I consider the following cases.
Case (1): The non-monetary equilibrium.
Case (2): The central bank sets $\mu = 1.82\%$ (i.e., the benchmark) and introduce CBDC with $R^c = 1\%$ and $\rho^c = 0.60$.
Case (3): Same as Case (2) except $R^c = 2\%$ and $\rho^c = 0.69$.
Case (4): Same as Case (2) except $R^c = 4\%$ and $\rho^c = 0.96$.
Note that given the choices of R^c's, 0.60, 0.69, and 0.96 are the highest values of ρ^c such that sellers who engage in money laundering prefer CBDC over cash. Proposition 7 shows that such values of ρ^c offer the highest aggregate welfare for any given R^c larger than the benchmark deposit rate (0.52%). Note also that in Case (1), as long as the equilibrium is non-monetary, the value of μ only affects nominal interest rates, while the real rates as well as other equilibrium outcomes are independent of μ.

The following table shows how equilibrium outcomes in the four cases compare with the monetary benchmark.

	Case (1): non-monetary	**Case (2): CBDC** $(R^c, \rho^c) = (1\%, 0.60)$	**Case (3): CBDC** $(R^c, \rho^c) = (2\%, 0.69)$	**Case (4): CBDC** $(R^c, \rho^c) = (4\%, 0.96)$
γ	-45.03 %	-12.19%	-26.06%	-50.12%
DM surplus	-0.30%	+0.16%	+0.45%	+0.84%
k	+0.55%	-0.32%	-0.96%	-2.20%
Total cost from money laundering	-46.21%	-10.35%	-22.04%	-43.97%
Aggregate welfare	+10.87%	+2.68%	+5.84%	+11.59%

Table 3: Percentage Change in Equilibrium Outcomes (relative to the monetary benchmark)

Electronic copy available at: https://ssrn.com/abstract=4394190

As predicted by theory, although money laundering is reduced in the non-monetary equilibrium, DM surplus is also lower compared to the monetary benchmark due to the decrease in bank deposit rate. Note that the same mechanism also explains why introducing a cash-like-high-anonymity CBDC decreases DM output and surplus (see Section 4.1). On the other hand, introducing high-interest-low-anonymity CBDC, such as the configurations considered in Table 3, increases DM surplus at the expense of capital investment. As discussed in Section 5.1, introducing such a CBDC increases the total supply of payment assets while also attracting sellers who do not launder money to accept CBDC. These two effects together mean that a decrease in money laundering will not lead to a decrease in deposit rate. However, a high-interest CBDC also raises all interest rates in the economy and reduces the demand for loans.

It is worth noting that a non-monetary equilibrium will not exist if there are other payment instruments that offer a level of anonymity similar to that of cash. For example, if domestic cash suffers from high inflation, sellers who engage in money laundering may switch to foreign cash or cryptocurrencies provided that they have lower inflation. Specifically, let ρ^f denote the probability that assets in a foreign cash are detected after a government audit. Let μ^f denote the inflation of the foreign cash. Then, sellers who engage in money laundering will choose foreign cash over domestic cash or bank deposits if

$$\frac{(1+\mu)/(1+R^d)}{1-\eta} > \frac{1+\mu^f}{1-\eta\rho^f} \quad \text{and} \quad \frac{(1+\mu)}{1-\eta\rho^m} > \frac{1+\mu^f}{1-\eta\rho^f}, \tag{7.2}$$

where η is the equilibrium audit probability. In such a case, increasing domestic inflation will not lead to a non-monetary equilibrium. It is therefore important to consider the privacy design of CBDC in the context of competing anonymous payment instruments such as cash.

8 Conclusion

The goal of this paper is to study the implications of money laundering for the optimal design of central bank digital currencies (CBDC). To accomplish this goal, I incorporate an audit game between the government and sellers who engage in money laundering into a general equilibrium model. As a benchmark, I assume only cash and bank deposits are available as payment instruments. Bank deposits have a higher return than cash, but it is easier to conceal cash from the government. I find that there exists a unique mixed strategy equilibrium, where some sellers choose to engage in money laundering, and the government audits sellers with probability strictly between zero and one.

When CBDC is introduced as a new payment instrument, the effect on money laundering depends crucially on the degree of anonymity of CBDC, which determines the probability of money laundering being discovered after a government audit if a seller's incomes are received in CBDC. I find that as long as CBDC offers less anonymity than cash, introducing CBDC will de-

Electronic copy available at: https://ssrn.com/abstract=4394190

crease money laundering. However, if CBDC is cash-like in the sense that it still offers relatively high anonymity but a low interest rate, then introducing CBDC will decrease the output from not only agents who launder money but also agents who do not. If CBDC is instead deposit-like in the sense that it offers low anonymity but a high interest rate, then introducing CBDC can increase aggregate welfare without lowering output. Furthermore, introducing deposit-like CBDC needs not increase the funding costs of private banks or decrease bank lending and investment.

To emphasize the mechanisms through which money laundering affects payment choice and CBDC design, I abstract from several important issues that could be addressed in future research. First, in this paper, there are no concerns of privacy other than money laundering. In practice, there are many reasons why individuals may want to hide their transactions from other individuals and the government (Kahn et al., 2020). The origins of such privacy concerns will likely determine the type(s) of transactions where CBDC will be accepted. This in turn can have important implications for money laundering and the central bank's revenue. Second, it has been argued that CBDC may help promote financial inclusion (Lagarde, 2018). Specifically, individuals may prefer cash not because they want to launder money, but because the transaction costs are lower compared to bank deposits and bank credit (Ait Lahcen and Gomis-Porqueras, 2021). If CBDC can be easy and cheap to use, it may promote transaction efficiency and improve welfare (Banet and Lebeau, 2022).

References

Agur, I., A. Ari, and G. Dell'Ariccia (2022). Designing central bank digital currencies. *Journal of Monetary Economics 125*, 62–79.

Ahnert, T., P. Hoffmann, and C. Monet (2022). The digital economy, privacy, and CBDC. ECB Working Paper.

Aït Lahcen, M. (2020). Informality, frictional markets and monetary policy. Working Paper.

Ait Lahcen, M. and P. Gomis-Porqueras (2021). A model of endogenous financial inclusion: implications for inequality and monetary policy. *Journal of Money, Credit and Banking 53*(5), 1175–1209.

Andolfatto, D. (2021). Assessing the impact of central bank digital currency on private banks. *The Economic Journal 131*(634), 525–540.

Andolfatto, D. and S. Williamson (2015). Scarcity of safe assets, inflation, and the policy trap. *Journal of Monetary Economics 73*, 70–92.

Aruoba, S. B. (2021). Institutions, tax evasion, and optimal policy. *Journal of Monetary Economics 118*, 212–229.

Bajaj, A. and N. Damodaran (2022). Consumer payment choice and the heterogeneous impact of india's demonetization. *Journal of Economic Dynamics and Control 137*, 104329.

Electronic copy available at: https://ssrn.com/abstract=4394190

Banet, J. and L. Lebeau (2022). Central bank digital currency: Financial inclusion vs. disintermediation. FRB of Dallas Working Paper.

Bank of Canada (2020). Contingency planning for a central bank digital currency. Retrieved from https://www.bankofcanada.ca/2020/02/contingency-planning-central-bank-digital-currency/.

Bank of Canada, European Central Bank, Bank of Japan, Sveriges Riksbank, Swiss National Bank, Bank of England, Board of Governors Federal Reserve System, and Bank for International Settlements (2020). Central bank digital currencies: foundational principles and core features. Retrieved from https://www.bis.org/publ/othp33.pdf.

Bank of England (2020). Central bank digital currency: Opportunities, challenges and design. Bank of England Discussion Paper. Retrieved from https://www.bankofengland.co.uk/-/media/boe/files/paper/2020/central-bank-digital-currency-opportunities-challenges-and-design.pdf.

Barone, R. and D. Masciandaro (2019). Cryptocurrency or usury? crime and alternative money laundering techniques. *European Journal of Law and Economics 47*, 233–254.

Barrdear, J. and M. Kumhof (2016). The macroeconomics of central bank issued digital currencies. Bank of England working paper.

Berentsen, A. and F. Schar (2018). The case for central bank electronic money and the non-case for central bank cryptocurrencies. Available at SSRN.

Boar, C., H. Holden, and A. Wadsworth (2020). Impending arrival–a sequel to the survey on central bank digital currency. BIS Paper.

Bordo, M. D. and A. T. Levin (2017). Central bank digital currency and the future of monetary policy. National Bureau of Economic Research.

Brunnermeier, M. K. and D. Niepelt (2019). On the equivalence of private and public money. *Journal of Monetary Economics 106*, 27–41.

Chiu, J., S. M. Davoodalhosseini, J. Hua Jiang, and Y. Zhu (2022). Market power and central bank digital currency: Theory and quantitative assessment. Journal of Political Economy, forthcoming.

Chiu, J. and T. Koeppl (2022). Paytech and the d (ata) n (etwork) a (ctivities) of bigtech platforms. Bank of Canada working paper.

Darbha, S. and R. Arora (2020). Privacy in CBDC technology. Technical report, Bank of Canada Staff Analytical Note 2020-9.

Davoodalhosseini, S. M. (2021). Central bank digital currency and monetary policy. *Journal of Economic Dynamics and Control*, 104150.

Dong, M. and S. X. Xiao (2021). Central bank digital currency: A corporate finance perspective. Working paper.

Drechsler, I., A. Savov, and P. Schnabl (2017). The deposits channel of monetary policy. *The Quarterly Journal of Economics 132*(4), 1819–1876.

Electronic copy available at: https://ssrn.com/abstract=4394190

ESCB (2019). Exploring anonymity in central bank digital currencies. European System of Central Banks (ESCB) Report in Focus Issue no 4, December. Retrieved from https://www.ecb.europa.eu/paym/intro/publications/pdf/ecb.mipinfocus191217.en.pdf.

Fernández-Villaverde, J., D. Sanches, L. Schilling, and H. Uhlig (2021). Central bank digital currency: central banking for all? *Review of Economic Dynamics 41*, 225–242.

Ferwerda, J., A. van Saase, B. Unger, and M. Getzner (2020). Estimating money laundering flows with a gravity model-based simulation. *Nature (Scientific reports) 10*(1), 18552.

Garcia, A., B. Lands, X. Liu, and J. Slive (2020). The potential effect of a central bank digital currency on deposit funding in canada. Bank of Canada Staff Analytical Note — 2020-15. Retrieved from https://www.bankofcanada.ca/wp-content/uploads/2020/07/san2020-15.pdf.

Garratt, R. and M. Lee (2021). Monetizing privacy with central bank digital currencies. Available at SSRN 3583949.

Garratt, R. J. and M. R. Van Oordt (2021). Privacy as a public good: a case for electronic cash. *Journal of Political Economy 129*(7), 2157–2180.

Gomis-Porqueras, P., A. Peralta-Alva, and C. Waller (2014). The shadow economy as an equilibrium outcome. *Journal of Economic Dynamics and Control 41*, 1–19.

Internal Revenue Service (2020). IRS data book. Retrieved from https://www.irs.gov/pub/irs-pdf/p55b.pdf.

Judson, R. (2017). The death of cash? not so fast: Demand for us currency at home and abroad, 1990-2016. Frankfurt a. M.: Deutsche Bundesbank.

Kahn, C. M., J. McAndrews, and W. Roberds (2005). Money is privacy. *International Economic Review 46*(2), 377–399.

Kahn, C. M., F. Rivadeneyra, and T.-N. Wong (2020). Should the central bank issue e-money? *Journal of Financial Market Infrastructures 8*(4), 1–22.

Kang, K.-Y. (2021). Digital currency and privacy. Available on SSRN.

Keister, T. and C. Monnet (2020). Central bank digital currency: Stability and information. Working paper.

Keister, T. and D. Sanches (2023). Should central banks issue digital currency? *The Review of Economic Studies 90*(1), 404–431.

Kwon, O., S. Lee, and J. Park (2022). Central bank digital currency, tax evasion, and inflation tax. *Economic Inquiry*.

Lagarde, C. (2018). Winds of change: The case for new digital currency. Speech by Christine Lagarde, IMF Managing Director, at Singapore Fintech Festival. Retrieved from https://www.imf.org/en/News/Articles/2018/11/13/sp111418-winds-of-change-the-case-for-new-digital-currency.

Lagos, R. and R. Wright (2005). A unified framework for monetary theory and policy analysis.

Electronic copy available at: https://ssrn.com/abstract=4394190

Journal of Political Economy 113(3), 463–484.

Lucas, R. E. (2000). Inflation and welfare. *Econometrica 68*(2), 247–274.

Lucas, R. E. and J. P. Nicolini (2015). On the stability of money demand. *Journal of Monetary Economics 73*, 48–65.

Masciandaro, D., E. Takats, and B. Unger (2007). *Black finance: the economics of money laundering*. Edward Elgar Publishing.

Powell, J. (2020). Speech at International Monetary Fund panel "cross-border payments—a vision for the future". Oct 19. Retrieved from `https://meetings.imf.org/en/2020/Annual/Schedule/2020/10/19/imf-cross-border-payments-a-vision-for-the-future`.

Ricks, M., J. Crawford, and L. Menand (2018). A public option for bank accounts (or central banking for all). Vanderbilt Law Research Paper.

Sanches, D. and S. Williamson (2010). Money and credit with limited commitment and theft. *Journal of Economic Theory 145*(4), 1525–1549.

Schilling, L., J. Fernández-Villaverde, and H. Uhlig (2020). Central bank digital currency: When price and bank stability collide. National Bureau of Economic Research.

Williamson, S. D. (2016). Scarce collateral, the term premium, and quantitative easing. *Journal of Economic Theory 164*, 136–165.

Williamson, S. D. (2019). Interest on reserves, interbank lending, and monetary policy. *Journal of Monetary Economics 101*, 14–30.

Williamson, S. D. (2021). Central bank digital currency and flight to safety. *Journal of Economic Dynamics and Control*, 104146.

Williamson, S. D. (2022). Central bank digital currency: welfare and policy implications. *Journal of Political Economy 130*(11), 2829–2861.

Electronic copy available at: https://ssrn.com/abstract=4394190

Appendix A Characterizations of CBDC Equilibria

A.1 Type-1 Equilibrium

In this type of equilibrium, $(1-\rho^c\eta)(1+\lambda P)(1+R^c) = 1-\rho^m\eta(1+\lambda P)$, so sellers who engage in money laundering are indifferent between cash and CBDC, while $R^c < R^d$, so sellers who do not engage in money laundering strictly prefer bank deposits. The equilibrium is characterized by the following.

(1) With probability $1-\gamma$, a seller does not engage in money laundering and posts (g, d), where g solves

$$u'(g) = \frac{1+\mu}{\beta(1+R^d)}, \tag{A.1}$$

and $d = \frac{1+R^d}{1+\mu}[\beta u(g) - \mathcal{S}]$.

(2) With probability $\iota\gamma$, a seller engages in money laundering and posts (q, c), where q solves

$$u'(q) = \frac{1+\mu}{\beta(1-\rho^c\eta)(1+\lambda P)(1+R^c)}, \tag{A.2}$$

and $c = \frac{1+R^c}{1+\mu}[\beta u(q) - \mathcal{S}]$.

(3) With probability $(1-\iota)\gamma$, a seller launders money and posts (q, m), where q solves (A.2), and m is

$$m = \frac{1}{1+\mu}[\beta u(q) - \mathcal{S}]. \tag{A.3}$$

(4) $\mathcal{S}$ is such that $d - g = 0$, and η is such that $(1-\rho^c\eta)(1+\lambda P)c - q = 0$.

(5) γ and ι satisfy $\iota\gamma\rho^c cN + (1-\iota)\gamma\rho^m mN = C$.

A.2 Type-2 Equilibrium

The equilibrium is characterized by the following.

(1) With probability $1-\gamma$, a seller does not engage in money laundering and posts (g, d), where g solves

$$u'(g) = \frac{1+\mu}{\beta(1+R^d)}, \tag{A.4}$$

and $d = \frac{1+R^d}{1+\mu}[\beta u(g) - \mathcal{S}]$.

(2) With probability γ, a seller engages in money laundering and posts (q, c), where q solves

$$u'(q) = \frac{1+\mu}{\beta(1-\rho^c\eta)(1+\lambda P)(1+R^c)}, \tag{A.5}$$

and $c = \frac{1+R^c}{1+\mu}[\beta u(q) - \mathcal{S}]$.

(3) $\mathcal{S}$ is such that $d - g = 0$, and η is such that $(1-\rho^c\eta)(1+\lambda P)c - q = 0$.

(4) γ solves $\gamma\rho^c cN = C$.

A.3 Type-3 Equilibrium

Let $R \equiv R^c = R^d$. The equilibrium is characterized by the following.

(1) With probability $1-\gamma$, a seller does not engage in money laundering and produce g, where g solves

$$u'(g) = \frac{1+\mu}{\beta(1+R)}. \tag{A.6}$$

Among sellers who do not engage in money laundering, a fraction ϵ accept bank deposits, while the rest

Electronic copy available at: https://ssrn.com/abstract=4394190

accept CBDC. Let a denote the payment received in bank deposits or CBDC by these sellers. Then a is

$$a = \frac{1+R}{1+\mu}[\beta u(g) - \mathcal{S}]. \tag{A.7}$$

(2) With probability γ, a seller engages in money laundering and posts (q, c), where q solves

$$u'(q) = \frac{1+\mu}{\beta(1-\rho^c\eta)(1+\lambda P)(1+R)}, \tag{A.8}$$

and $c = \frac{1+R}{1+\mu}[\beta u(q) - \mathcal{S}]$.
(3) $\mathcal{S}$ is such that $d - g = 0$, and η is such that $(1-\rho^c\eta)(1+\lambda P)c - q = 0$.
(4) γ solves $\gamma\rho^c cN = C$.
(5) The demand for government bonds from bankers is given by $\bar{b} = \frac{\epsilon(1-\gamma)g}{1-\theta} - f'(k)k$.
(6) The total demand for CBDC is given by $\bar{c} = (1-\gamma)(1-\epsilon)g + \frac{(1+\lambda)C}{\rho^c N}$.
(7) ϵ is such that $\bar{b} + \bar{c} = \mathcal{D}$.

A.4 Type-4 Equilibrium

Let $R \equiv R^c = R^d$. The equilibrium is characterized by the following.
(1) With probability $1-\gamma$, a seller does not engage in money laundering and produce g, where g solves

$$u'(g) = \frac{1+\mu}{\beta(1+R)}. \tag{A.9}$$

Among sellers who do not engage in money laundering, a fraction ϵ accept bank deposits, while the rest accept CBDC. Let a denote the payment received in bank deposits or CBDC. Then a is given by

$$a = \frac{1+R}{1+\mu}[\beta u(g) - \mathcal{S}]. \tag{A.10}$$

(2) With probability γ, a seller engages in money laundering and posts (q, m), where q solves

$$u'(q) = \frac{1+\mu}{\beta(1-\rho^m\eta^0)}, \tag{A.11}$$

and $m = \frac{1}{1+\mu}[\beta u(q) - \mathcal{S}]$.
(3) $\mathcal{S}$ is such that $d - g = 0$, and η^0 is such that $(1-\rho^m\eta(1+\lambda P))m - q = 0$.
(4) γ solves $\gamma\rho^m mN = C$.
(5) The demand for government bonds from bankers is given by $\bar{b} = \frac{\epsilon(1-\gamma)g}{1-\theta} - f'(k)k$.
(6) The total demand for CBDC is given by $\bar{c} = (1-\gamma)(1-\epsilon)g$.
(7) The total demand for cash is given by $\bar{m} = \frac{(1+\lambda)C}{\rho^m N}$.
(8) ϵ is such that $\bar{m} + \bar{b} + \bar{c} = \mathcal{D}$.

Appendix B Omitted Proofs

Proof of Proposition 1: First, consider the sellers who choose not to engage in money laundering. Based on the solutions from the main text, we have

$$u'(g) = \frac{1+\mu}{\beta(1+R^d)}, \tag{B.1}$$

and

$$d = \frac{1+R^d}{1+\mu}[\beta u(q^h) - \mathcal{S}]. \tag{B.2}$$

Electronic copy available at: https://ssrn.com/abstract=4394190

Next, consider sellers who engage in money laundering. Assume that they accept only cash. Then, we have

$$u'(q) = \frac{1+\mu}{\beta[(1-\eta\rho^m)(1+\lambda P)]}, \tag{B.3}$$

and

$$m = \frac{\beta u(q) - \mathcal{S}}{1+\mu}. \tag{B.4}$$

For such sellers to prefer cash, it must be that

$$\frac{(1+\mu)/(1+R^d)}{(1-\eta)(1+\lambda P)} > \frac{1+\mu}{(1-\eta\rho^m)(1+\lambda P)}. \tag{B.5}$$

In other words, for any R^d and η, there exists $\overline{\rho^m}$ such that as long as $\rho^m < \overline{\rho^m}$, the sellers prefer cash. Next, for sellers to be indifferent between engaging or not engaging in money laundering, η must satisfy

$$d - g = [(1-\eta\rho^m)(1+\lambda P)]m - q. \tag{B.6}$$

It is straightforward to see that there exists $\underline{\rho^m}$ such that if $\rho^m > \underline{\rho^m}$, there exists $\eta < 1$ that solves (B.6).

Now, I show that sellers' surplus must be driven to zero in 'equilibrium. To see why, first note that in equilibrium there must not exist a submarket where sellers are strictly better off while buyers are at least as well off. Second, because there are more sellers than buyers, it must be that $n(q,p) < 1$ for some (q,p). If sellers' surplus is positive, then consider a submarket with (q,p,n') where $n' > n(q,p)$. In this submarket, buyers are as well off but sellers are strictly better off, which is a contradiction. In conclusion, we have $d - g = [(1-\eta\rho^m)(1+\lambda P)]m - q = 0$.

Finally, for the government to be indifferent between auditing or not auditing a seller, the following condition must be satisfied:

$$C = \gamma\rho^m mN. \tag{B.7}$$

Hence, it is clear that there exists $\overline{C}$ such that if $C < \overline{C}$, then $\gamma \in (0,1)$. □

Proof of Proposition 2: First, given R^d and μ, g solves

$$u'(g) = \frac{1+\mu}{\beta(1+R^d)}. \tag{B.8}$$

Next, η is such that

$$\frac{(1-\eta\rho^m)(1+\lambda P)}{1+\mu}\left[\beta u(q) - \beta u(g) + \frac{(1+\mu)g}{1+R^d}\right] - q = 0. \tag{B.9}$$

And $\gamma = \frac{C}{\rho^m mN}$. The total demand for cash is given by

$$\bar{m} = \gamma m(1+\lambda) = \frac{(1+\lambda)C}{\rho^m N}. \tag{B.10}$$

Note that $\frac{1+\mu}{1+R^b} = \beta\theta + \frac{(1-\theta)(1+\mu)}{1+R^d}$. Then, R^b is such that

$$f'(k) = \frac{1+R^b}{1+\mu}, \tag{B.11}$$

$$(1-\theta)(k+\bar{b}) = (1-\gamma)g. \tag{B.12}$$

Note that (B.12) defines $\bar{b}$ as a function of R^b (and hence R^d). Then,

$$\bar{m} + \bar{b} = \frac{(1-\gamma)g}{1-\theta} - f'(k)k + \frac{(1+\lambda)C}{\rho^m N}. \tag{B.13}$$

Then $\bar{m}+\bar{b}$ is decreasing in γ. Note that γ is decreasing in m, and m is increasing in $\mathcal{S}$. To see the latter

Electronic copy available at: https://ssrn.com/abstract=4394190

point, first note that

$$\beta u(q) - \frac{(1+\mu)q}{(1-\eta\rho^m)(1+\lambda P)} = \mathcal{S} \tag{B.14}$$

and $m = \frac{q}{(1-\eta\rho^m)(1+\lambda P)}$. Define $G(A) = Aq$ where q solves $u'(q) = A(1+\mu)/\beta$. Then

$$G'(A) = q\left[1 + \frac{u'(q)}{qu''(q)}\right] < 0, \tag{B.15}$$

because by assumption $\frac{u'(q)}{qu''(q)} < -1$. Hence, m is decreasing in $\frac{1}{(1-\eta\rho^m)(1+\lambda P)}$, which in turn is decreasing in $\mathcal{S}$. Furthermore, g and $\mathcal{S}$ are increasing in R^d. Hence, γ is decreasing in R^d. If $kf'(k)$ is increasing in k, then $kf'(k)$ is decreasing in R^d. In conclusion, $\bar{m} + \bar{b}$ is strictly increasing in R^d. Hence, a unique equilibrium exists where $\bar{m} + \bar{b} = \mathcal{D}$. □

Proof of Proposition 3: First, consider an increase in ρ^m. Assume that $\frac{1+R^d}{1+\mu}$ is unchanged. Then $\mathcal{S}$ is unchanged. This means that q is unchanged as well, Since

$$u'(q) = \frac{1+\mu}{\beta[(1-\eta\rho^m)(1+\lambda P)]}, \tag{B.16}$$

η must be smaller. Because $m = \frac{q}{(1-\eta\rho^m)(1+\lambda P)}$, the demand for cash is unchanged. Since $\gamma = \frac{C}{\rho^m mN}$, fewer sellers engage in money laundering. Now, I prove that $g/(1-\theta) > (1+\lambda)m$. Note that because buyers are indifferent between engaging or not engaging in money laundering, we have

$$\beta u(g) - \frac{1+\mu}{1+R^d}g = \beta u(q) - \frac{1+\mu}{(1-\eta\rho^m)(1+\lambda P)}q. \tag{B.17}$$

It is clear that $g = q$. Hence, it must be that $\frac{g}{1+R^d} = \frac{q}{(1-\eta\rho^m)(1+\lambda P)} = m$. Since $R^d \geq 0$, we have $g/(1-\theta) > m(1+\lambda)$. Hence, $\bar{m} + \bar{b}$ is decreasing in γ, and if $\frac{1+R^d}{1+\mu}$ is unchanged, $\bar{m} + \bar{b} > \mathcal{D}$. Because $\bar{m} + \bar{b}$ is increasing in R^b, in equilibrium it must be that $\frac{1+R^d}{1+\mu}$ is lower. Then, $\mathcal{S}$ and g must be smaller in equilibrium, and so is q. Now, suppose γ is unchanged after the increase in ρ^m. Because g is smaller and $\frac{1+R^d}{1+\mu}k$ is larger, (B.12) suggests that $\bar{b}$ is smaller. Hence, in equilibrium, γ must be smaller.

Next, consider an increase in μ and assume that $\frac{1+R^d}{1+\mu}$ is unchanged. Then $\mathcal{S}$ is unchanged. This means that q is unchanged as well, Since $u'(q) = \frac{1+\mu}{\beta[(1-\eta\rho^m)(1+\lambda P)]}$, and μ is higher, η must decrease. Because $m = \frac{q}{(1-\eta\rho^m)(1+\lambda P)}$, the demand for cash is lower. Since $\gamma = \frac{C}{\rho^m mN}$, more sellers engage in money laundering. Since $\bar{m} + \bar{b}$ is decreasing in γ, if μ is higher and $\frac{1+R^d}{1+\mu}$ is unchanged, then $\bar{m} + \bar{b} < \mathcal{D}$. Because $\bar{m} + \bar{b}$ is increasing in R^b, in equilibrium it must be that $\frac{1+R^d}{1+\mu}$ is larger. To conclude, as μ increases, $\frac{1+R^d}{1+\mu}$ increases and η decreases. Hence, there must exist a μ' such that if $\mu > \mu'$, sellers who engage in money laundering will prefer to accept bank deposits even if $\rho^m = 0$. In such a case, there does not exist a monetary equilibrium where cash is accepted as payment. □

Proof of Proposition 4: First, given R^d and μ, g solves

$$u'(g) = \frac{1+\mu}{\beta(1+R^d)}. \tag{B.18}$$

Next, η is such that

$$u(g) - \frac{1+\mu}{\beta(1+R)}g = u(q) - \frac{1+\mu}{\beta(1-\rho^c\eta)(1+\lambda P)(1+R)}q. \tag{B.19}$$

Let $c = q/(1-\rho^c\eta)(1+\lambda P)$. Then, γ is such that

$$\gamma\rho^c cN = C. \tag{B.20}$$

Electronic copy available at: https://ssrn.com/abstract=4394190

The demand for CBDC is given by

$$\bar{c} = \gamma(1+\lambda)c = \frac{(1+\lambda)C}{\rho^c N}. \tag{B.21}$$

Note that $\frac{1+\mu}{1+R^b} = \beta\theta + \frac{(1-\theta)(1+\mu)}{1+R^d}$. Then, R^b is such that

$$f'(k) = \frac{1+R^b}{1+\mu}, \tag{B.22}$$

$$(1-\theta)(k+\bar{b}) = (1-\gamma)g. \tag{B.23}$$

Then,

$$\bar{c} + \bar{b} = \frac{(1-\gamma)g}{1-\theta} - f'(k)k + \frac{(1+\lambda)C}{\rho^c N}. \tag{B.24}$$

Consider an increase in R^c. If R^d is unchanged, then g and q are unchanged. But then $(1-\rho^c\eta)(1+\lambda P)$ will decrease, which means c will increase and γ will decrease. Hence, R^d must decrease, which means g and q will decrease. Consider an increase in ρ^c. If R^d is unchanged, then g, q, and c will be unchanged, but γ will decrease. Hence, R^d must decrease, which means g and q will decrease. □

Proof of Proposition 5: Define $R \equiv R^d = R^c$. First, given R and μ, g solves

$$u'(g) = \frac{1+\mu}{\beta(1+R)}. \tag{B.25}$$

Next, η is such that

$$u(g) - \frac{1+\mu}{\beta(1+R)}g = u(q) - \frac{1+\mu}{\beta(1-\rho^c\eta)(1+\lambda P)(1+R)}q, \tag{B.26}$$

Let $c = q/((1-\rho^c\eta)(1+\lambda P))$. Then, γ is such that

$$\gamma\rho^c cN = C. \tag{B.27}$$

The demand for government bonds is given by

$$f'(k) = \frac{1+R^b}{1+\mu}, \tag{B.28}$$

$$(1-\theta)(k+\bar{b}) = (1-\gamma)g, \tag{B.29}$$

where R^b is given by $\frac{1+\mu}{1+R^b} = \beta\theta + \frac{(1-\theta)(1+\mu)}{1+R}$. Then, we have

$$\bar{c} + \bar{b} = \frac{(1+\lambda)C}{\rho^c N} + (1-\epsilon)(1-\gamma)g + \frac{\epsilon(1-\gamma)g}{1-\theta} - f'(k)k. \tag{B.30}$$

Now, consider an increase in R. Then, g and q will increase. In addition, c will increase because $c = q/((1-\rho^c\eta)(1+\lambda P)) = \beta(1+R)qu'(q)/(1+\mu)$. This means γ and ϵ will decrease. Finally, consider an increase in ρ^c. If R^c is unchanged, then g, q, and c will be unchanged. However, this means that γ will decrease. □

Proof of Proposition 6: First, given R^c and μ, g solves

$$u'(g) = \frac{1+\mu}{\beta(1+R)}. \tag{B.31}$$

Next, η is such that

$$u(g) - \frac{1+\mu}{\beta(1+R^c)}g = u(q) - \frac{1+\mu}{\beta[(1-\rho^m\eta)(1+\lambda P)]}q. \tag{B.32}$$

Let $m = q/[(1-\rho^m\eta)(1+\lambda P)]$. And γ is such that

$$\gamma\rho^m mN = C. \tag{B.33}$$

Electronic copy available at: https://ssrn.com/abstract=4394190

The demand for government bonds is given by

$$f'(k) = \frac{1+R^b}{1+\mu}, \tag{B.34}$$

$$(1-\theta)(k+\bar{b}) = \epsilon(1-\gamma)g. \tag{B.35}$$

Then,

$$\bar{m} + \bar{c} + \bar{b} = \frac{(1+\lambda)C}{\rho^m N} + (1-\epsilon)(1-\gamma)g + \frac{\epsilon(1-\gamma)g}{1-\theta} - f'(k)k. \tag{B.36}$$

Now, consider an increase in R^c while μ is held constant. Then g and q will increase. In addition, m will increase because $m = q/[(1-\rho^m\eta)(1+\lambda P)] = \beta q u'(q)/(1+\mu)$. Hence, γ will decrease. Since the demand for government liabilities increases, ϵ must decrease. □

Proof of Proposition 7: First, note that from the FOC of the government's audit problem, we have $C = \rho^c\gamma\mathcal{P}N$. Hence, aggregate welfare can be rewritten as

$$\mathcal{W} = (1-\gamma)[u(g)-g] + \gamma[u(q)-q] + \beta f(k) - k - \gamma\mathcal{P}N. \tag{B.37}$$

Note also that from the audit problem, we have $\gamma = \frac{C}{\rho\mathcal{P}N}$, where $\rho = \rho^c$ and $\mathcal{P} = c$ if CBDC is accepted by sellers who engage in money laundering. Hence,

$$\gamma\mathcal{P}N = \frac{C}{\rho^c}. \tag{B.38}$$

For any given R^c, I only need to show that when $\rho^c = \rho^c(R^c)$, γ is larger in a type-4 equilibrium than in a type-3 equilibrium. Note that on the boundary of type-3 and type-4 equilibria, it must be that

$$\frac{1+\mu}{(1-\rho^c\eta)(1+\lambda P)(1+R^c)} = \frac{1+\mu}{(1-\rho^m\eta)(1+\lambda P)}. \tag{B.39}$$

Hence, η is equal in type-3 and type-4 equilibria. Next, note that $c = q/[(1-\rho^c\eta)(1+\lambda P)] > m = q/[(1-\rho^m\eta)(1+\lambda P)]$. In addition, in a type-4 equilibriumγ is given by

$$\gamma\rho^m mN = C. \tag{B.40}$$

In a type-3 equilibrium, γ is given by

$$\gamma\rho^c cN = C. \tag{B.41}$$

Hence, γ is larger in a type-4 equilibrium than in a type-3 equilibrium. □

Proof of Proposition 8: From Proposition 7, we know that if $(R^c, \rho^c) = (R^{bench}, \rho^c(R^{bench}))$, q, g and k in a type-3 equilibrium are identical to their counterparts in the benchmark. However, γ is strictly lower. This means that the total cost from money laundering is lower. Hence, the aggregate welfare is strictly higher in a type-3 equilibrium than in the benchmark equilibrium. Next, consider cash-like CBDC. As long as R^c and ρ^c are close to point $(0, \rho^m)$, then the change in equilibrium outcomes will be small compared to the benchmark. Hence, aggregate welfare is higher with deposit-like CBDC than with cash-like CBDC. □

Appendix C Supplement to Quantitative Analysis

In this appendix, I recalibrate the model with $\tilde{N} = 7$ (Figure 11-13) and $P = 0.2$ (Figure 14-16), while keeping other targets unchanged. I report the quantitative results with the new parameters values below.

Electronic copy available at: https://ssrn.com/abstract=4394190

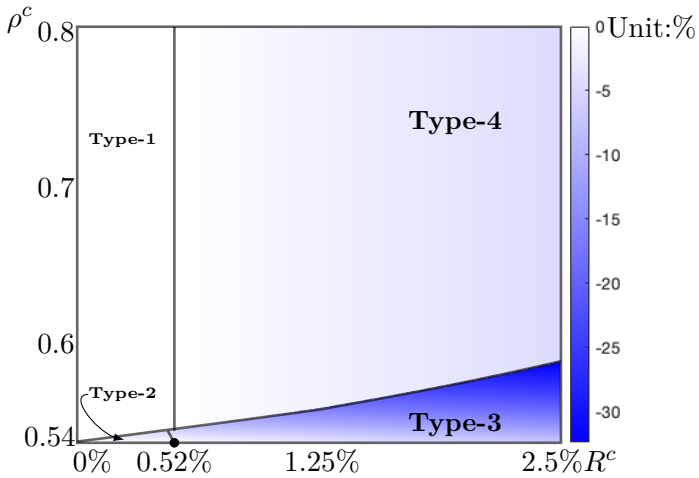


Figure 11: Percentage Change in Money Laundering (γ) ($\tilde{N} = 7$)

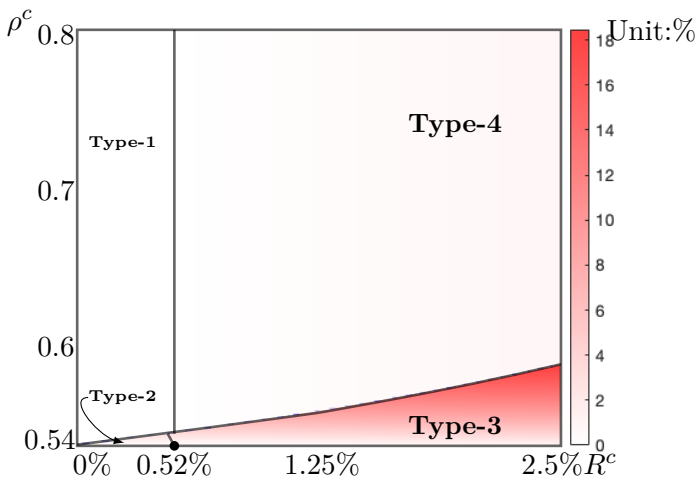


Figure 12: Percentage Change in Aggregate Welfare ($\tilde{N} = 7$)

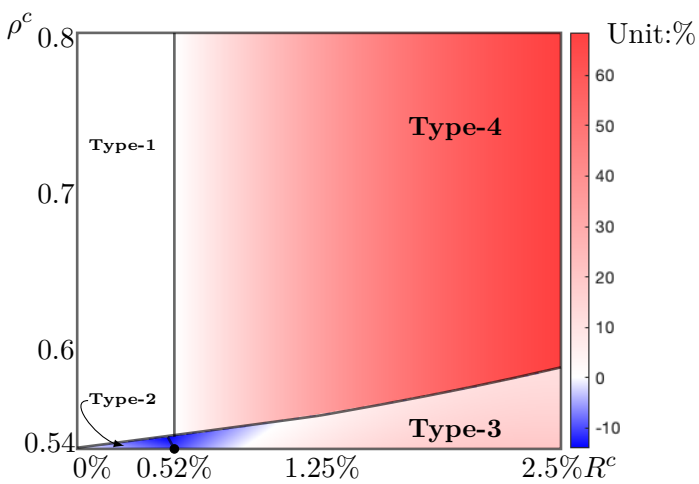


Figure 13: Percentage Change in Central Bank Revenue ($\tilde{N} = 7$)

Electronic copy available at: https://ssrn.com/abstract=4394190

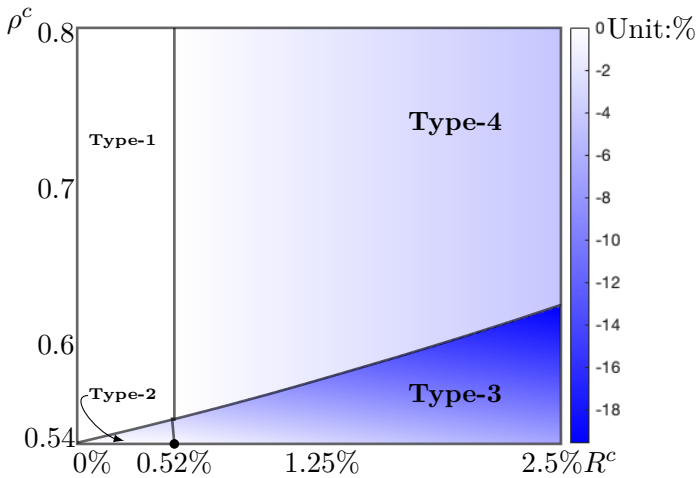


Figure 14: Percentage Change in Money Laundering (γ) ($P = 0.2$)

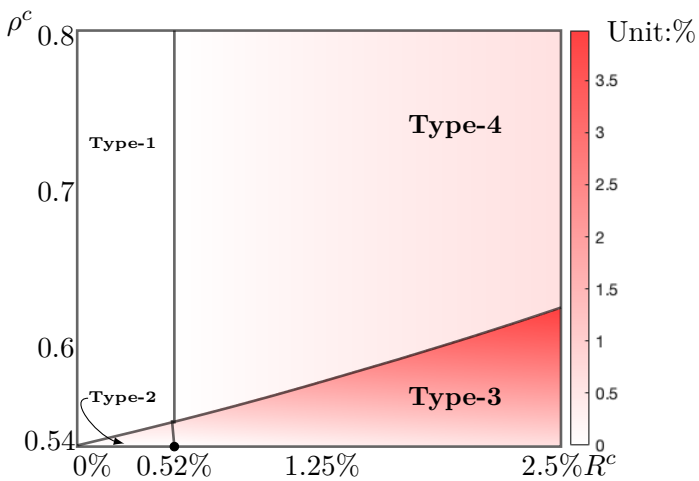


Figure 15: Percentage Change in Aggregate Welfare ($P = 0.2$)

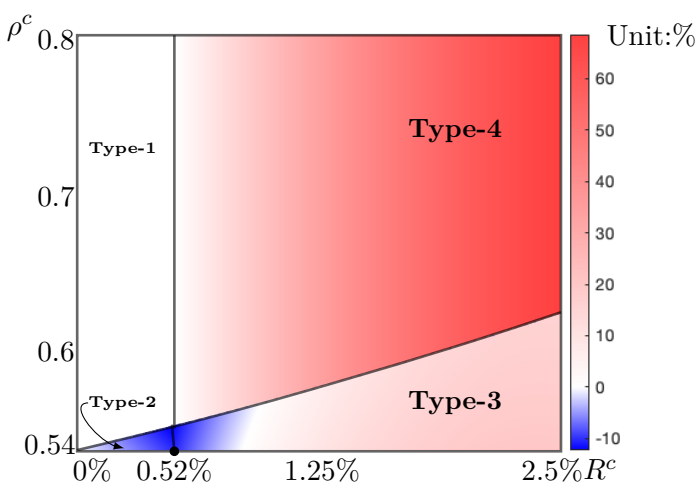


Figure 16: Percentage Change in Central Bank Revenue ($P = 0.2$)

Electronic copy available at: https://ssrn.com/abstract=4394190

Appendix D Extensions

D.1 Markets for Illegal Goods and Money Laundering Services

In this section, I model explicitly the markets for money laundering services and illegal goods to endogenize the source of illegal income and $\mathcal{P}$. Suppose that in addition to buyers and sellers of the DM good, there is also measure one of drug dealers and measure one of drug addicts, and both types of agents are only active in the CM. Drug dealers cannot work in the CM, but they can turn one unit of the CM good into $a > 1$ units of drugs. I normalize the cost of drug production to zero. Drug dealers also derive linear utility from consuming the CM good. Drug addicts have access to the same linear technology buyers use to produce the CM good. Their utility function is given by

$$U(X_t) + D_t - H_t, \tag{D.1}$$

where X_t is the consumption of the CM good, D_t is the consumption of drugs, and H_t is the labor supplied in the CM. I assume that $U'(x) > 0$, $U''(x) < 0$, $U'(\infty) = 0$, and $U'(0) = \infty$.

It is straightforward to show that the optimal X_t is given by $U'(X_t) = 1$. Now, consider how D_t is determined. I assume that drug addicts and dealers are randomly and bilaterally matched in a drug market that opens in each CM, and drug dealers make take-it-or-leave-it offers to drug addicts. I assume that the sale and purchase of CM good are monitored in the CM, so if drug addicts pay directly with the CM good, drug trade will be discovered and prevented by the government. Similarly, drug dealers cannot use the payments (cash, deposits, and/or CBDC) that they receive directly from buyers to purchase the CM good. For drug trade to happen, drug dealers must send the payments to DM sellers who can disguise them as part of legal DM income, after which the payments are "laundered" and can be used to purchase the CM good. I summarize the process in Figure 17.

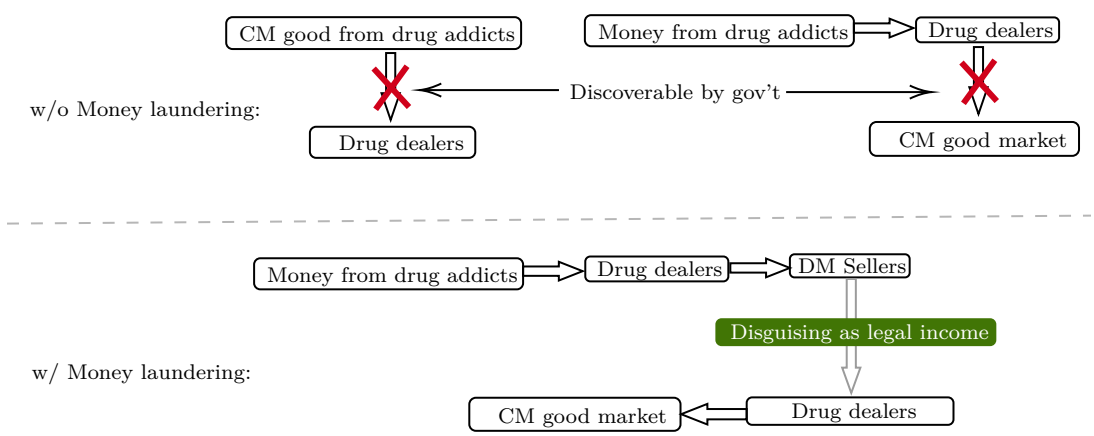


Figure 17: Drug Trade With or Without Money Laundering

Before I describe the market for money laundering services, recall that DM sellers post in the CM the offers for the next DM, which consist of the quantity of DM good they offer to produce, x, and the payment that they expect to receive, $p \equiv (m, c, d)$. Buyers observe all offers before they decide which seller to visit. Let $n(x, p)$ denote the buyer-to-seller ratio in each submarket. I assume a buyer is matched with a seller in the CM with probability $\min\{1, 1/n(x, p)\}$, and a seller is matched with a buyer in the CM with probability $\min\{1, n(x, p)\}$. However, the production and consumption of the DM good as well as the exchange of payments still take place in the DM.

DM sellers who are matched with buyers can choose to enter a market for money laundering services that opens before the drug market. I assume that sellers are randomly and bilaterally matched with drug dealers. The price of money laundering services is determined via proportional bargaining, with DM sellers' bargaining power being Θ. For ease of exposition, assume for the moment that a seller only

Electronic copy available at: https://ssrn.com/abstract=4394190

accepts cash (i.e., $c = d = 0$). Recall that by assumption, for m units of legal cash income, the seller can launder λm units of illegal income. Now, let D denote the quantity of drugs the dealer will produce. Since the drug dealer makes a take-it-or-leave-it offer in the following drug market, a drug addict's expected surplus must be zero:

$$(1 - \rho^m \mathbb{E}(\eta))D - (1 + \mu)\lambda m = 0. \tag{D.2}$$

Note that $\mathbb{E}(\eta)$ is the expected audit probability, so $(1-\rho^m\mathbb{E}(\eta))$ is the probability that laundered income is *not* confiscated, in which case drugs will be produced in the next CM. Recall that Q units of drugs require Q/a units of CM good to produce. Hence, the total surplus from drug trade is

$$\left[1 - \frac{1 + \mu}{a(1 - \rho^m \mathbb{E}(\eta))}\right] \lambda m. \tag{D.3}$$

Since the seller's bargaining power is Θ, her payoff per unit of money laundered is given by

$$\mathcal{P} = \Theta \left[1 - \frac{1 + \mu}{a(1 - \rho^m \mathbb{E}(\eta))}\right]. \tag{D.4}$$

It is worth noting that $\mathcal{P}$ is no longer a constant, and it is decreasing in $\mathbb{E}(\eta)$.

Now, substitute $\mathcal{P}$ into the solution to sellers' problem (see Section 3.1). We have

$$m = \frac{\beta u(q) - \mathcal{S}}{1 + \mu}, \tag{D.5}$$

where $\mathcal{S}$ is buyer's surplus, and q solves

$$\beta u'(q) = \frac{1 + \mu}{[1 + \lambda(\Theta - \mathcal{C})][1 - \mathbb{E}(\eta)\rho^m] - \frac{\lambda\Theta(1+\mu)}{a}}. \tag{D.6}$$

If the seller only accepts bank deposits, then the solution is given by

$$d = \frac{\beta u(q) - \mathcal{S}}{(1 + \mu)/(1 + R^d)}, \tag{D.7}$$

where q solves

$$\beta u'(q) = \frac{(1 + \mu)/(1 + R^d)}{[1 + \lambda(\Theta - \mathcal{C})][1 - \mathbb{E}(\eta)] - \frac{\lambda\Theta(1+\mu)/(1+R^d)}{a}}. \tag{D.8}$$

The seller will only accept cash (i.e., $d = 0$) if

$$\frac{1 + \mu}{[1 + \lambda(\Theta - \mathcal{C})][1 - \mathbb{E}(\eta)\rho^m] - \frac{\lambda\Theta(1+\mu)}{a}} > \frac{(1 + \mu)/(1 + R^d)}{[1 + \lambda(\Theta - \mathcal{C})][1 - \mathbb{E}(\eta)] - \frac{\lambda\Theta(1+\mu)/(1+R^d)}{a}}. \tag{D.9}$$

The problem for sellers who do not engage in money laundering as well as the government's audit problem remain unchanged. In what follows, I prove that Propositions 2 and 3 still hold. Since the theoretical results regarding CBDC are based on these two propositions, they also remain unchanged.

I. Existence and uniqueness of a general equilibrium

From the proof of Proposition 2, we know that given R^d and μ, g solves

$$u'(g) = \frac{1 + \mu}{\beta(1 + R^d)}. \tag{D.10}$$

Second, η is such that

$$\frac{[1 + \lambda(\Theta - \mathcal{C})][1 - \eta\rho^m] - \frac{\lambda\Theta(1+\mu)}{a}}{1 + \mu} \left[\beta u(q) - \beta u(g) + \frac{(1 + \mu)g}{1 + R^d}\right] - q = 0. \tag{D.11}$$

Third, γ is given by the FOC of the government's audit problem:

$$C = \gamma \rho^m m N. \tag{D.12}$$

Electronic copy available at: https://ssrn.com/abstract=4394190

The total demand for cash is given by

$$\bar{m} = \gamma(1+\lambda)m = \frac{(1+\lambda)C}{\rho^m N}. \tag{D.13}$$

Note that $\frac{1+\mu}{1+R^b} = \beta\theta + \frac{(1-\theta)(1+\mu)}{1+R^d}$. Then, R^b is such that $f'(k) = \frac{1+R^b}{1+\mu}$ and

$$(1-\theta)(k+\bar{b}) = (1-\gamma)g. \tag{D.14}$$

We have

$$\bar{m} + \bar{b} = \frac{(1-\gamma)g}{1-\theta} + \frac{(1+\lambda)C}{\rho^m N} - f'(k)k. \tag{D.15}$$

Then $\bar{m}+\bar{b}$ is decreasing in γ. Note that γ is decreasing in m, and m is increasing in $\mathcal{S}$. To see the latter point, first note that

$$\beta u(q) - \frac{(1+\mu)q}{[1+\lambda(\Theta-\mathcal{C})][1-\mathbb{E}(\eta)\rho^m] - \frac{\lambda\Theta(1+\mu)}{a}} = \mathcal{S} \tag{D.16}$$

and $m = \frac{q}{[1+\lambda(\Theta-\mathcal{C})][1-\mathbb{E}(\eta)\rho^m]-\frac{\lambda\Theta(1+\mu)}{a}}$. Define $G(A) = Aq$ where q solves $u'(q) = A(1+\mu)/\beta$. Then

$$G'(A) = q\left[1 + \frac{u'(q)}{qu''(q)}\right] < 0, \tag{D.17}$$

because by assumption $\frac{u'(q)}{qu''(q)} < -1$. Hence, m is decreasing in $\frac{1}{[1+\lambda(\Theta-\mathcal{C})][1-\mathbb{E}(\eta)\rho^m]-\frac{\lambda\Theta(1+\mu)}{a}}$, which in turn is decreasing in $\mathcal{S}$. Furthermore, g and $\mathcal{S}$ are increasing in R^d. Hence, γ is decreasing in R^d. If $kf'(k)$ is increasing in k, then $kf'(k)$ is decreasing in R^d. In conclusion, $\bar{m}+\bar{b}$ is strictly increasing in R^d. Hence, a unique equilibrium exists where $\bar{m}+\bar{b} = \mathcal{D}$.

II. The effects of ρ^m and μ on the general equilibrium

First, consider an increase in ρ^m. Assume that $\frac{1+R^d}{1+\mu}$ is unchanged. Then $\mathcal{S}$ is unchanged. This means that q is unchanged as well, Since

$$u'(q) = \frac{1+\mu}{[1+\lambda(\Theta-\mathcal{C})][1-\eta\rho^m] - \frac{\lambda\Theta(1+\mu)}{a}}, \tag{D.18}$$

η must be smaller. Because $m = \frac{q}{(1-\eta\rho^m)(1+\lambda P)}$, the demand for cash remains unchanged. Since $\gamma = \frac{C}{\rho^m mN}$, fewer sellers engage in money laundering. Note that because buyers are indifferent between engaging or not engaging in money laundering, we have $g = q$. Hence, it must be that $\frac{d}{1+R^d} = m$. Since $R^d \geq 0$, we have $g/(1-\theta) > m(1+\lambda) > m$. Hence, if $\frac{1+R^d}{1+\mu}$ is unchanged, $\bar{m}+\bar{b} > \mathcal{D}$. Because $\bar{m}+\bar{b}$ is increasing in R^d, in equilibrium it must be that $\frac{1+R^d}{1+\mu}$ is lower. Then, $\mathcal{S}$ and g must be smaller in equilibrium, and so is q. Now, suppose γ is unchanged compared to before. Then, because g is smaller and $\frac{1+R^d}{1+\mu}k$ is larger, (B.12) suggests that $\bar{b}$ is smaller. This means that in equilibrium, γ must be smaller.

Next, consider an increase in μ and assume that $\frac{1+R^d}{1+\mu}$ is unchanged. Then $\mathcal{S}$ is unchanged. This means that q is unchanged as well, Since $u'(q) = \frac{1+\mu}{[1+\lambda(\Theta-\mathcal{C})][1-\eta\rho^m]-\frac{\lambda\Theta(1+\mu)}{a}}$, and μ is higher, η must decrease. Because

$$m = \frac{q}{[1+\lambda(\Theta-\mathcal{C})][1-\eta\rho^m] - \frac{\lambda\Theta(1+\mu)}{a}},$$

the demand for cash is lower. Since $\gamma = \frac{C}{\rho^m mN}$, more sellers engage in money laundering. Since $\bar{m}+\bar{b}$ is decreasing in γ, if μ is higher and $\frac{1+R^d}{1+\mu}$ is unchanged, then $\bar{m}+\bar{b} < \mathcal{D}$. Because $\bar{m}+\bar{b}$ is increasing in R^b, in equilibrium it must be that $\frac{1+R^d}{1+\mu}$ is larger. To conclude, as μ increases, $\frac{1+R^d}{1+\mu}$ increases and η decreases. Hence, there must exist a μ' such that if $\mu > \mu'$, sellers who engage in money laundering will prefer to accept bank deposits even if $\rho^m = 0$. In such a case, there does not exist a monetary equilibrium

Electronic copy available at: https://ssrn.com/abstract=4394190

where cash is accepted as payment.

To conclude, the qualitative effects of ρ^m and μ on the general equilibrium are identical to those found in Proposition 3. Hence, main results in this paper remain changed.

D.2 General Audit Cost Function

In this section, I consider a general functional form for the audit cost. For ease of exposition, I use the benchmark model in Section 3 as an example, but the conclusions also apply to the version with markets for illegal goods and money laundering services (see Appendix D.1). In what follows, I prove that the results in Propositions 2 and 3 hold under Assumption D.1. Since the theoretical results regarding CBDC are based on these two propositions, they remain unchanged under the general audit cost function.

Assumption D.1 (1) *The total cost of audits is given by $\tilde{C}(A)$, where A represents the measure of audits conducted, and it satisfies $\tilde{C}'(A) > 0$ and $C''(A) > 0$.* (2) *$[(1-\eta\rho^m)(1+\lambda P)]\tilde{C}'(\alpha\eta)$ is decreasing in η.*

Under audit cost function $\tilde{C}(A)$, the government's audit problem becomes

$$\min_{\eta} -\alpha\gamma \min\{1, n(q,m,d)\}\{(1-\eta)[\mathbf{1}(m>0)mN + \mathbf{1}(d>0)dN] + \eta\mathbf{1}(m>0)(1-\rho^m)mN\} - \{\tilde{C}(\eta\alpha)\}$$

I. Existence and uniqueness of a general equilibrium under $\tilde{C}(A)$

From the proof of Proposition 2, we know that given R^d and μ, g solves

$$u'(g) = \frac{1+\mu}{\beta(1+R^d)}. \tag{D.19}$$

Second, η is such that

$$\frac{(1-\eta\rho^m)(1+\lambda P)}{1+\mu}\left[\beta u(q) - \beta u(g) + \frac{(1+\mu)g}{1+R^d}\right] - q = 0. \tag{D.20}$$

Third, γ is given by the FOC of the government's audit problem:

$$\alpha\tilde{C}'(\eta\alpha) = \gamma\rho^m mN. \tag{D.21}$$

The total demand for cash is given by

$$\bar{m} = \gamma(1+\lambda)m = \frac{\alpha(1+\lambda)\tilde{C}'(\eta\alpha)}{\rho^m N}. \tag{D.22}$$

Note that $\frac{1+\mu}{1+R^b} = \beta\theta + \frac{(1-\theta)(1+\mu)}{1+R^d}$. Then, R^b is such that $f'(k) = \frac{1+R^b}{1+\mu}$ and

$$(1-\theta)(k+\bar{b}) = (1-\gamma)g. \tag{D.23}$$

Then,

$$\bar{m} + \bar{b} = \frac{(1-\gamma)g}{1-\theta} + \frac{\alpha(1+\lambda)\tilde{C}'(\eta\alpha)}{\rho^m N} - f'(k)k. \tag{D.24}$$

Now, I prove that $\bar{m} + \bar{b}$ is strictly increasing in R^d. First, note that g and $\mathcal{S}$ are increasing in R^d. This means that η must be decreasing in R^d so that (D.20) holds. Define $G(A) = q$ where q solves $u'(q) = A(1+\mu)/\beta$. Then

$$G'(A) = q\left[1 + \frac{u'(q)}{qu''(q)}\right] < 0, \tag{D.25}$$

because by assumption $\frac{u'(q)}{qu''(q)} < -1$. Hence, m is decreasing in $\frac{1}{(1-\eta\rho^m)(1+\lambda P)}$ and increasing in R^d. From (D.21), since $C''(.) > 0$, we can conclude that γ is decreasing in R^d. Now, define

$$H = \frac{(1-\gamma)g}{1-\theta} + \frac{\alpha(1+\lambda)\tilde{C}'(\alpha\eta)}{\rho^m N}. \tag{D.26}$$

Electronic copy available at: https://ssrn.com/abstract=4394190

We have

$$\begin{aligned}\frac{\mathrm{d}H}{\mathrm{d}R^d} &= (1-\gamma)\frac{\mathrm{d}g}{\mathrm{d}R^d} - \frac{g}{1-\theta}\frac{\mathrm{d}\gamma}{\mathrm{d}R^d} + \frac{\mathrm{d}\left(\frac{\alpha(1+\lambda)\tilde{C}'(\alpha\eta)}{\rho^m N}\right)}{\mathrm{d}R^d} \\ &= (1-\gamma)\frac{\mathrm{d}g}{\mathrm{d}R^d} - \frac{g}{1-\theta}\frac{\mathrm{d}\left(\frac{\alpha(1+\lambda)\tilde{C}'(\alpha\eta)}{\rho^m mN}\right)}{\mathrm{d}R^d} + \frac{\mathrm{d}\left(\frac{\alpha(1+\lambda)\tilde{C}'(\alpha\eta)}{\rho^m N}\right)}{\mathrm{d}R^d} \\ &= (1-\gamma)\frac{\mathrm{d}g}{\mathrm{d}R^d} - \left[\frac{g}{m(1-\theta)} - 1\right]\frac{\mathrm{d}\left(\frac{\alpha(1+\lambda)\tilde{C}'(\alpha\eta)}{\rho^m N}\right)}{\mathrm{d}R^d} - \frac{g}{1-\theta}\frac{\alpha(1+\lambda)\tilde{C}'(\alpha\eta)}{\rho^m N}\frac{\mathrm{d}\left(\frac{1}{m}\right)}{\mathrm{d}R^d}.\end{aligned}$$

Recall that both m and g are increasing in R^d, and that $\frac{\mathrm{d}\left(\frac{\alpha(1+\lambda)\tilde{C}'(\alpha\eta)}{\rho^m N}\right)}{\mathrm{d}R^d} < 0$. To show that $\frac{\mathrm{d}H}{\mathrm{d}R^d} > 0$, I only need to show that $\frac{g}{m(1-\theta)} > 1$. Note that because buyers are indifferent between engaging or not engaging in money laundering, we have $g = q$. Hence, it must be that $\frac{g}{1+R^d} = m$. Since $R^d \geq 0$, we have $g/(1-\theta) > m$. Thus, we can conclude that $\frac{\mathrm{d}H}{\mathrm{d}R^d} > 0$. Since $kf'(k)$ is increasing in k, then $kf'(k)$ is decreasing in R^d. Hence, $\bar{m} + \bar{b}$ is strictly increasing in R^d. In conclusion, a unique equilibrium exists where $\bar{m} + \bar{b} = \mathcal{D}$.

II. The effects of ρ^m and μ on the general equilibrium

First, consider an increase in ρ^m. Assume that $\frac{1+R^d}{1+\mu}$ is unchanged. Then $\mathcal{S}$ is unchanged. This means that q is unchanged as well, Since

$$u'(q) = \frac{1+\mu}{(1-\eta\rho^m)(1+\lambda P)}, \tag{D.27}$$

η must be smaller. Because $m = \frac{q}{(1-\eta\rho^m)(1+\lambda P)}$, the demand for cash remains unchanged. Since $\gamma = \frac{\alpha\tilde{C}'(\alpha\eta)}{\rho^m mN}$, fewer sellers engage in money laundering. Note that because buyers are indifferent between engaging or not engaging in money laundering, we have $g = q$. Hence, it must be that $\frac{d}{1+R^d} = m$. Since $R^d \geq 0$, we have $g/(1-\theta) > m(1+\lambda)$. Hence, if $\frac{1+R^d}{1+\mu}$ is unchanged, $\bar{m} + \bar{b} > \mathcal{D}$. Because $\bar{m} + \bar{b}$ is increasing in R^d, in equilibrium it must be that $\frac{1+R^d}{1+\mu}$ is lower. Then, $\mathcal{S}$ and g must be smaller in equilibrium, and so is q. Now, suppose γ is unchanged compared to before. Then, because g is smaller and $\frac{1+R^d}{1+\mu}k$ is larger, (B.12) suggests that $\bar{b}$ is smaller. This means that in equilibrium, γ must be smaller.

Next, consider an increase in μ and assume that $\frac{1+R^d}{1+\mu}$ is unchanged. Then $\mathcal{S}$ is unchanged. This means that q is unchanged as well, Since $u'(q) = \frac{1+\mu}{\beta[(1-\eta\rho^m)(1+\lambda P)]}$, and μ is higher, η must decrease. Because $m = \frac{q}{(1-\eta\rho^m)(1+\lambda P)}$, the demand for cash is lower. Since

$$\gamma = \frac{\alpha\tilde{C}'(\alpha\eta)}{\rho^m mN} = \frac{[(1-\eta\rho^m)(1+\lambda P)]\alpha\tilde{C}'(\alpha\eta)}{q\rho^m N}, \tag{D.28}$$

and $[(1-\eta\rho^m)(1+\lambda P)]\tilde{C}'(\alpha\eta)$ is decreasing in η by assumption, γ is larger. Since $\bar{m} + \bar{b}$ is decreasing in γ, if μ is higher and $\frac{1+R^d}{1+\mu}$ is unchanged, then $\bar{m} + \bar{b} < \mathcal{D}$. Because $\bar{m} + \bar{b}$ is increasing in R^b, in equilibrium it must be that $\frac{1+R^d}{1+\mu}$ is larger. To conclude, as μ increases, $\frac{1+R^d}{1+\mu}$ increases and η decreases. Hence, there must exist a μ' such that if $\mu > \mu'$, sellers who engage in money laundering will prefer to accept bank deposits even if $\rho^m = 0$. In such a case, there does not exist a monetary equilibrium where cash is accepted as payment.

To conclude, the qualitative effects of ρ^m and μ on the general equilibrium are identical to those found in Proposition 3. Hence, the main results in this paper remain changed.

Electronic copy available at: https://ssrn.com/abstract=4394190